Master Level Thesis

Energy Efficient Built Environment

A comparative LCA of CLT and Glulam reuse with wooden reversible connections in the HasleTre project

Master thesis 15 credits, 2026
Energy Efficient Built Environment

Author:
Gaëlle De Longueville and Camille De Boeck

Supervisor:
Jonn Are Myrhen

Examiner:
Tony Svensson

Course Code: EG3020

Examination date: 4th of June 2026

HÖGSKOLAN DALARNA

MASTER THESIS

Dalarna University
Energy Engineering

Vid Högskolan Dalarna finns möjlighet att publicera examensarbetet i fulltext i DiVA. Publiceringen sker Open Access, vilket innebär att arbetet blir fritt tillgängligt att läsa och ladda ned på nätet. Därmed ökar spridningen och synligheten av examensarbetet.

Open Access är på väg att bli norm för att sprida vetenskaplig information på nätet. Högskolan Dalarna rekommenderar såväl forskare som studenter att publicera sina arbeten Open Access.

Jag/vi medger publicering i fulltext (öppet tillgänglig på nätet, Open Access):

Ja ☒ Nej ☐

Abstract

Considering the proportional increase of embodied carbon in the built environment, this thesis evaluates the climate mitigation potential of material circularity in mass timber construction. The primary objective is to quantify, via Life Cycle Assessment (LCA), the Global Warming Potential (GWP) reduction of a circular timber case study compared to a linear baseline.

Using a mixed-methods approach, the study integrates qualitative insights from an industry Focus Group Discussion (FGD) with quantitative modelling in One Click LCA. Designed for Disassembly (DfD), the HasleTre office building in Oslo features a modular design and reversible connections to enable a theoretical structural reusability rate approaching 100 %. This case study is evaluated against a linear reference project with a reuse baseline estimated at approximately 70 %. By isolating the Cross-Laminated Timber (CLT) panels, a multi-cycle evaluation examines the influence of different allocation methods on the panels environmental profile across three lifecycles.

The comparative LCA reveals a marginal GWP difference between the buildings due to nearly identical material volumes, though the circular model potentially retains more long-term biogenic carbon. Crucially, the results uncover a module D paradox under current standards, where immediate wood incineration receives higher environmental credits than material reuse. While direct GWP reductions from the all-wood joints are negligible, they offer an opportunity to act as the primary strategic enablers for clean mechanical disassembly and high-value structural recovery. Multi-cycle CLT evaluation results vary widely from 39 to 127 tonnes of CO_2e, emphasising a high sensitivity to accounting frameworks. Furthermore, extending a single building lifespan to 100 years proves more environmentally effective (560 tonnes of CO_2e) than a two-cycle rebuilding strategy where the primary structure is reused (900 tonnes of CO_2e).

The thesis concludes that while reversible connections are essential tactical tools to secure material loop integrity, maximising initial building longevity remains the primary environmental objective. However, significant uncertainties and knowledge gaps remain regarding the actual long-term structural durability and behaviour of CLT panels during repeated disassembly and reuse phases. To accurately reflect and reward the environmental performance of timber circularity, current linear accounting practices must evolve into standardised multi-cycle frameworks.

Acknowledgment

With this present text, we would like to express our gratitude to everyone who contributed to this master thesis.

First, we want to thank Jonn Are Myrhen, our supervisor for his constant enthusiasm, his valuable guidance, and the generous time he dedicated to our work. We are also grateful to Ambrose Dodoo and Felix Quarcoo for their advice and support as part of the research project.

Our thanks also go to Moritz Groba from OsloTre for his time, his valuable insights, and his willingness to share his expertise. His commitment to bringing timber construction to such a high level of sustainability has been a significant inspiration for this work.

We would also like to thank the participants of the Focus Group Discussion for sharing their professional expertise, as well as the teachers and students of the Energy Efficient Built Environment programme for providing a supportive and stimulating environment for our studies at Dalarna University.

Finally, we want to thank our friends for the productive study sessions and much-needed breaks, and our families for their constant support in our studies and personal projects.

Contents

1 Introduction 1
1.1 Background 2
1.2 Aims and Objectives 4
1.3 Methodology 4
1.4 Limitations 5
1.4.1. Geography 5
1.4.2. Subjectivity of Qualitative Data 5
1.4.3. Modelling Constraints 5
1.4.4. Indicator Scope: GWP 5

2 Literature Review and Theory 6
2.1 Timber Construction 6
2.2 CLT and Glulam 6
2.2.1. Structural Systems 7
2.2.2. Connections 9
2.2.3. CLT Buildings in LCA 9
2.2.4. End-of-life Practices of CLT Today 10
2.2.5. Glulam Buildings in LCA 10
2.2.6. End-of-life Practices of Glulam Today 11
2.3 Design Strategies for Circularity 12
2.3.1. Design for Disassembly (DfD) 12
2.3.2. Design for Adaptability (DfA) 13
2.3.3. DfD and DfA in Timber Construction 13
2.4 Life Cycle Assessment (LCA) 13
2.4.1. Accounting for Reuse 14
2.4.2. Accounting for Biogenic Carbon 15
2.5 Research Gap 16

3 Case Study Description 17

4 Materials and Methods 21
4.1 Focus Group Discussion 21
4.2 Scope 22
4.3 Comparative LCA 22
4.3.1. Scope, Boundary and Data 22
4.3.2. Life Cycle Analysis 24
4.3.3. Model of the HasleTre Building (HT Circular) 24
4.3.4. Model of the HasleTre Reference Building (HT Reference) 25
4.4 Multicycle Evaluation of CLT Panels 25
4.4.1. Scope and Boundary 25
4.4.2. Life Cycle Analysis 26
4.4.3. Cut-off Method (100:0) 27
4.4.4. End-of-life Method (0:100) 27
4.4.5. Distributed Method 28
4.5 Sensitivity Analysis 28
4.5.1. Lifespan 29
4.5.2. End-of-Life Scenarios 29

5 Results 30
5.1 Focus Group Discussion 30
5.2 Comparative LCA (HT Circular – HT Reference) 32

5.2.1. Contribution of the Stages 32
5.2.2. Contribution by Material for the Production Stage 35
5.3 Multi-Cycle Evaluation of CLT Panels 37
5.4 Sensitivity Analysis 40
5.4.1. Lifespan 40
5.4.2. End-of-Life Scenarios 42
6 Discussion 46
6.1 Discussion of the Results 46
6.1.1. Materials 46
6.1.2. Connections 47
6.1.3. Module D Paradox 48
6.1.4. Extending the Lifespan 49
6.1.5. Frameworks, Certification, and Ecosystem Readiness 49
6.2 Discussion of the Method 50
7 Conclusions 52
7.1 Applicability of the Study 52
7.2 Future Work 53
References 54
Appendix A 59
Appendix B 62
Appendix C 64
Appendix D 66

Abbreviations

Abbreviation	Description
CE	Circular Economy
CLT	Cross-Laminated Timber
CO_2	Carbon dioxide
DfA	Design for Adaptability
DfD	Design for Disassembly
EoL	End-of-Life
EPD	Environmental Product Declaration
FGD	Focus Group Discussion
GHG	Greenhouse Gas
Glulam	Glued Laminated Timber
GWP	Global Warming Potential
LCA	Life Cycle Assessment
BBR	Boverket's Building Regulations

Nomenclature

Symbol	Description	Unit
I_C	Environmental impact of the construction	kg CO_2e
I_D	Environmental impact of the disposal	kg CO_2e
I_P	Environmental impact of the production	kg CO_2e
I_R	Environmental impact of the reuse (loads)	kg CO_2e
R_1	Share of reused materials used in the building	/
R_2	Share of material reusable at the end of life of the building	/
GWP	Global warming potential	kg CO_2e
I	Total environmental impact	kg CO_2e

1 Introduction

Faced with today's environmental challenges, the construction sector, which accounts for a significant proportion of greenhouse gas (GHG) emissions and waste generation in Europe, is being forced to rethink its practices. Whilst traditional sustainability efforts in the Nordic countries have focused on reducing operational energy, the sector is now confronted with the hidden impact of embodied carbon. To address this, the implementation of circular strategies has become an important strategy. These strategies are particularly effective when combined with the use of wood-based materials, such as Cross-Laminated Timber (CLT) or Glued Laminated Timber (Glulam), which offer the possibility to store biogenic carbon.

This thesis explores the transition from a linear construction model to a circular one by evaluating the HasleTre project, in Oslo, Norway. This pioneering example demonstrates how timber construction can serve as a temporary carbon sink through innovative design that facilitates the disassembly and reuse of materials.

To evaluate this potential, a methodology in four steps is followed, allowing a comprehensive evaluation using both qualitative insights and quantitative environmental data. The first step establishes the research context with a Focus Group Discussion (FGD) with industry experts. Secondly, a comparative Life Cycle Assessment (LCA) between the circular HasleTre building and its non-circular equivalent is conducted. Then the CLT panels are isolated for a multi-cycle evaluation to test how different allocation methods influence the results. Finally, sensitivity analyses evaluate the impact of variables such as building longevity and alternative disposal scenarios, ensuring the robustness of the results.

1.1 Background

Today, global warming has become a primary concern for the global population, making the reduction of GHG emissions an urgent priority at the centre of many international political discussions. The 2015 Paris Agreement commits the 194 countries involved to pursue efforts to limit the temperature increase to 1.5°C above pre-industrial levels by the end of this century [1]. To achieve this goal, GHG emissions must be reduced by 43 % by 2030.

The construction sector is a critical factor in this challenge. It is currently responsible for 40 % of the European Union (EU) total energy use and 36 % of its GHG emissions [2]. Furthermore, the sector production of waste is substantial: in 2022, 38 % of all waste produced in Europe was linked to construction [3]. Regionally, construction activities generated approximately 13 million tonnes of waste in Sweden (including mineral waste) [4], and 2.7 million tonnes in Norway (excluding mineral waste)[5]. Consequently, the built environment is identified as a key area for improvement to reduce the overall environmental impact.

While construction waste is generated throughout a building entire life cycle, including during the design phase, the End-of-Life (EoL) phase is the most significant contributor [6]. In fact, 50 % of the total waste produced during the life cycle of a building occurs during this final phase. This high proportion can be explained by the fact that most materials are discarded or downcycled at the end of their life because they lack potential for reuse or, even when such potential exists, they are rarely reused.

Given these challenges, it is becoming essential to find solutions to reduce GHG emissions and waste in construction.

One response to these challenges lies in the transition from a linear economic model to a circular one. The European Parliament defines Circular Economy (CE) as:

> "A model of production and consumption, which involves sharing, leasing, reusing, repairing, refurbishing and recycling existing materials and products as long as possible. In this way, the life cycle of products is extended" [7].

Despite its potential benefits, implementation of circular economy remains limited. In Sweden, only 3.4 % of the resources used are cycled back into the economy after their initial use [8], while in Norway, the figure is even lower at 2.4 % [9].

The shift toward a circular economy is driven by various European strategic frameworks. The EU Circular Economy Action Plan, published in 2015, identified the construction and demolition sector as a strategic priority [10]. This emphasises the importance of reuse and recycling in preserving the value of materials. Furthermore, the Waste Framework Directive (Directive 2008/98/EC, 2008) establishes a clear hierarchy for waste prevention and management: prevention, preparation for reuse, recycling, other recovery (including energy recovery) and, as last resort, disposal [11]. More recently, the European Taxonomy has introduced a classification system to define sustainable economic activities [12], and the Ecodesign for Sustainable Products Regulation (ESPR) sets out performance and information requirements. It promotes eco-construction and reuse through tools such as the Digital Product Passport (DPP) for materials, acting as a digital passport that tracks the material data on its manufacturing, usage, and recycling potential. [13].

These initiatives aim to encourage circularity in the construction sector, where various strategies have emerged. Among these, reuse is a particularly promising alternative for reducing both waste

and the carbon footprint associated with the production of new materials. To facilitate this transition, Design for Disassembly (DfD) and Adaptability (DfA) are essential technical solutions. Adopting these strategies can significantly increase the potential for reuse compared to standard practices, reaching up to 85 % of building components [14].

However, achieving climate goals also requires a fundamental shift in material selection. Historically, climate action in Scandinavia has focused on reducing operational energy demand during the building use phase, leading to high-performance insulation and efficient thermal systems [15]. However, as the energy grid decarbonises and operational efficiency improves, the embodied carbon of building materials has become proportionally more influential. It is projected to increase even more in the coming years, going from 25 % to 49 % of the carbon contributions within the building sector, as represented on Figure 1.1 [16]. This is highlighted in research by Malmqvist et al., indicating that the material production accounts for a major part (56 %) of a Swedish building Global Warming Potential (GWP). This surpasses the impact of the operational phase, which was found to constitute 39 % of the life-cycle emissions [17].

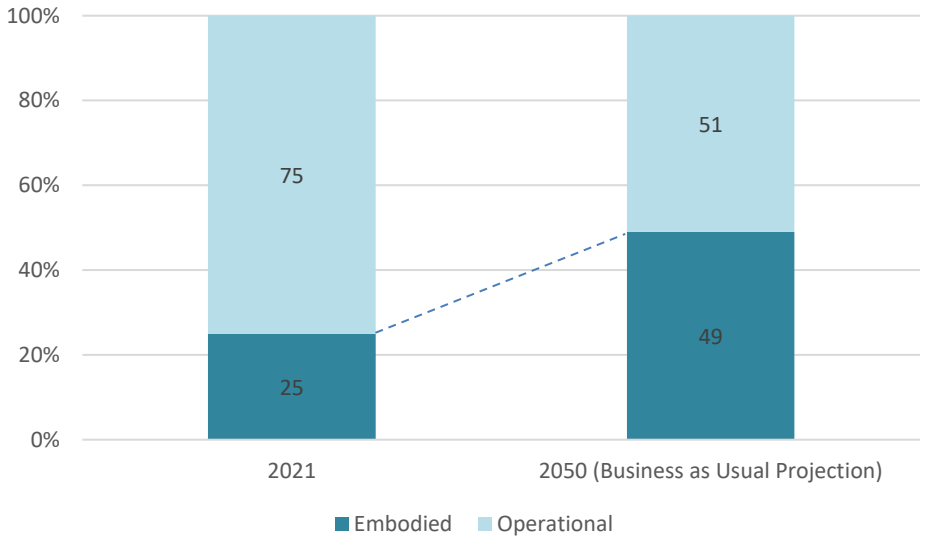


Figure 1.1: Projected carbon contributions from embodied and operational carbon within the building sector, reprinted from United Nations Environment Programme and Yale Center for Ecosystems + Architecture [16]

As said, the construction industry currently acts as a major source of emissions. This is partially due to the processing, manufacturing and transportation of building material, particularly from concrete and steel production, which are responsible for a large share of global emissions [18]. Addressing this issue presents an opportunity to achieve global climate goals. By utilising timber sourced from sustainable forestry, urban environments can function as "second forests", providing long-term carbon storage within our cities. Indeed, as forests become less reliable for carbon storage due to disturbances linked to climate change, and given that current fossil fuel combustion rates have largely exceeded the natural sequestration capacities, the built environment offers an opportunity [19]. Construction materials with high carbon storage capacity and low embodied carbon emissions, such as wood based materials, enable the development of a carbon pool by locking it into a physical structure while minimising the emissions typically associated with conventional material production.

Furthermore, reports of Intergovernmental Panel on Climate Change and various individual LCA studies have demonstrated the superior environmental performance of timber buildings, which are characterised by lower life-cycle primary energy use and decreased GHG emissions, compared to

traditional buildings materials [20]. This global potential is particularly well suited to the Nordic countries, where abundant forest resources and technical expertise have enabled the emergence of CLT and Glulam to facilitate the large-scale development of circular and carbon-neutral construction [21], [22].

By combining circular design strategies with low carbon bio-based materials, the construction sector can transition from a linear and high emission industry to a more circular and carbon-sequestering component of the global climate solution.

1.2 Aims and Objectives

Based on the broader context established in the preceding Background, this section defines the aims and objectives of the study. The primary aim of this thesis is to evaluate the environmental impacts of reusing CLT and Glulam elements with wooden reversible connections in the HasleTre project through a comparative LCA. This project is a pioneering, five-storey timber office building in Oslo, Norway, specifically designed for disassembly and structural material reuse. Focusing on CLT panels, this research aims to investigate further how different EoL allocation methods influence their environmental profile over multiple lifespans.

To achieve this aim, the following specific objectives have been established:

- Identify specific circular solutions by discussing with industry experts to gain insights, used to narrow down the circular solutions evaluated.
- Quantify and compare GWP of circular designed strategies by comparing a baseline model of the case study against a conventional reference project of identical size and function to evaluate the potential benefits of adopting a circular construction model.
- Assess the CLT panels over three consecutive life cycles by applying and comparing different EoL allocation methods to understand how environmental burdens are distributed across multiple uses.
- Evaluate the impact of variables by changing key parameters such as total building lifespan and different EoL scenarios to understand the impact on the overall environmental benefits.

1.3 Methodology

To achieve the established objectives, this section outlines the mixed-methods research framework. The study combines qualitative industry insights with quantitative environmental results to achieve the objectives. The method follows four main phases:

- Qualitative data collection through a FGD and parameter definition

The first stage involves a FGD with several industry experts (including an architect, contractors, and a structural engineer). Various circular solutions in the timber construction are discussed and a Wooclap [23] ranking system is used to evaluate the feasibility of various technical solutions. Based on those insights, some solutions (specifically the use of wooden reversible connections) are translated into quantitative material inventories and input parameters for the LCA.

- Comparative LCA

A comparative LCA is conducted using the One Click LCA software to evaluate two distinct building models and quantify their respective GWP, defined as follows:

- Circular scenario: model based on the real-life HasleTre building, incorporating wooden reversible connections and a high reusability rate, defined here as the inherent potential of materials to be reused at the end of life of the building.
- Reference project: building having the same function and size as the case study, with metal fasteners and consequently a lower reusability rate.

This assessment includes the primary structure and the building envelope components but excludes building service systems and furniture.

- Multi-cycle evaluation of CLT panels

An LCA isolates CLT panels, which have a high reuse proportion thanks to the DfD strategies. They are modelled across three consecutive life cycles with three different EoL allocation method (100:0, 0:100 and distributed).

- Sensitivity analysis

A sensitivity analysis is conducted, testing the impact of the building lifespan and different EoL scenarios.

1.4 Limitations

The findings of this research must be interpreted within the study defined boundaries. The following limitations regarding geography, data subjectivity of the FGD, modelling constraints, and the exclusive focus on GWP are identified in this section to help contextualising the results. A critical examination of these factors and their implication are addressed in the Discussion.

1.4.1. Geography

This study is specifically concentrated on a case study located in Norway with the geographic boundary extended to the Nordic region of Norway and Sweden. Because the electric grids in these countries is largely decarbonised [24], the proportional significance of embodied carbon is increased. Furthermore, in this region, timber construction is common. This specialised context may not be directly applicable to other regions with different electricity mix and material availability.

1.4.2. Subjectivity of Qualitative Data

The qualitative insights are derived from a single FGD consisting of five industry experts. Given that these participants are already convinced of the benefits of the circular economy, their views may be somewhat biased and may not be representative of the construction sector, which tends to be more conservative. Consequently, the ranking and discussions are subjective and serve as recommendations rather than a universal industry consensus.

1.4.3. Modelling Constraints

Assumptions are made for the modelling of the HasleTre Building. The primary structure and building envelope were considered but not the services systems or furniture to focus on the transition of linear to circular for the structure.

1.4.4. Indicator Scope: GWP

The quantitative analysis is limited to the GWP. By isolating this indicator, a potential shift of burden may not be identified [25].

2 Literature Review and Theory

Having established the main research framework, and objectives centred on the environmental benefits of material circularity in multi-storey wood building with a focus on CLT panels, it is necessary to situate this study in a broader academic context. In this way, a comprehensive analysis of the technical concept that governs sustainable timber construction in the Nordic region will provide guidance for the study and help establish the scope of this research.

To define a robust theoretical foundation for evaluating circular solutions in mass timber construction, this literature review explores the environmental potential and structural characteristics of CLT and Glulam, the principles of DfD and DFA, and the current methodological challenges in quantifying biogenic carbon flows and the benefits of material reuse.

2.1 Timber Construction

Literature agrees that timber buildings have a lower environmental impact than traditional construction [20]. Recent research by Amiri et al. quantifies the potential for European buildings to serve as a significant carbon sink over the 2020-2040 period by analysing 50 case studies [18]. The researchers categorised wooden structures into three capacity tiers and developed four distinct construction scenarios based on market adoption rates. Their findings reveal that increasing timber construction could sequester between 1 and 55 Mt of carbon dioxide (CO_2) annually, a volume capable of offsetting up to 47 % of the total emissions generated by the European cement industry. This highlights the importance of promoting timber construction in Europe in the coming years. This also applies to the Swedish context, where the increased use of timber construction appears to be an effective approach to reducing carbon emissions of the Swedish building stock, according to the study from Peñaloza et al. on long-term climate mitigation strategies for new buildings [26].

However, it is essential to specify that the environmental performance of timber buildings is influenced by the accounting of biogenic carbon flows [26]. Indeed, carbon flows are cyclical as the carbon sequestered during tree growth is returned to the atmosphere at the end of the product life cycle, establishing wood products as temporary carbon storage. Consequently, strategies that extend the service life of timber elements, through reuse and disassembly, are important to maximise the duration of the carbon storage. Several strategies for accounting for biogenic carbon and reuse are set out in Section 2.4.2.

In addition to this necessity for circularity, the timber sector in the Nordic region, specifically Norway, Sweden, and Finland has grown over the past few decades [21]. This is due to three primary drivers: the abundant availability of local productive forest resources, a long-standing technical expertise in wood processing derived from traditional small-scale construction, and an increasing demand for carbon-neutral and circular building solutions. In this context, the emergence of CLT and Glulam has played a decisive role in extending the technical and structural boundaries of mass timber, thereby enabling greater capacity for larger-scale projects.

2.2 CLT and Glulam

The shift toward larger-scale and circular mass timber project is facilitated by the implementation of CLT and Glulam. As urbanisation continues with an estimated 60 % of the global population residing in cities by 2030, construction activity necessitates resource-efficient building methods to preserve natural capital [27]. In this context, mass timber construction offers an opportunity to mitigate the embodied carbon of the built environment while meeting the demands of urban density.

2.2.1. Structural Systems

There are three categories of mass timber structural systems, mainly linked with steel fasteners, that are commonly used (Figure 2.1)[27]. First, the post-and-beam framing utilises a skeleton of Glulam posts and beams to support timber decking such as CLT. The interior walls are non-load bearing because the primary loads are carried by the frame. Consequently, it provides high architectural flexibility.

Secondly, the hybrid system combines mass timber floor and roof panels with wood or steel light-frame bearing walls. This system is suitable for mid-rise multi-family buildings, as light frame walls facilitate integration of HVAC systems while providing reliable acoustic and fire performance.

Finally, for the mass timber bearing walls, the CLT panels serve as both the floor and the primary load-bearing walls. This system leverages the two-way spanning capabilities of CLT and can even replace traditional concrete cores. It is suitable for high-rise timber construction due to the structural efficiency and high load-bearing capacity. CLT is therefore of interest to promote low carbon buildings [28].

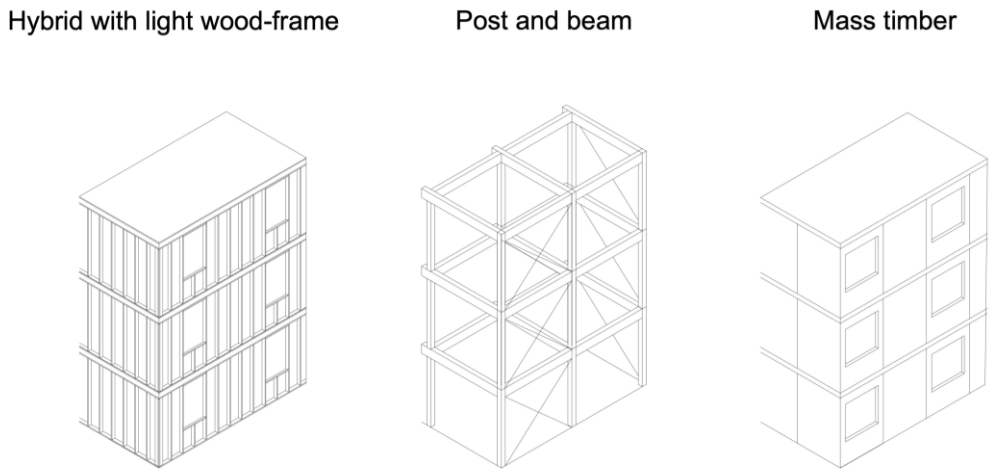


Figure 2.1: Three different building typologies, reprinted from Chapagain and Crovella [27].

CLT panels are manufactured using timber laminations glued together at 90-degree angles, as illustrated on Figure 2.2. CLT structures need prefabrication, as the element are too massive to be cut on the construction site [29]. They are transported there in order to be assembled with connectors.

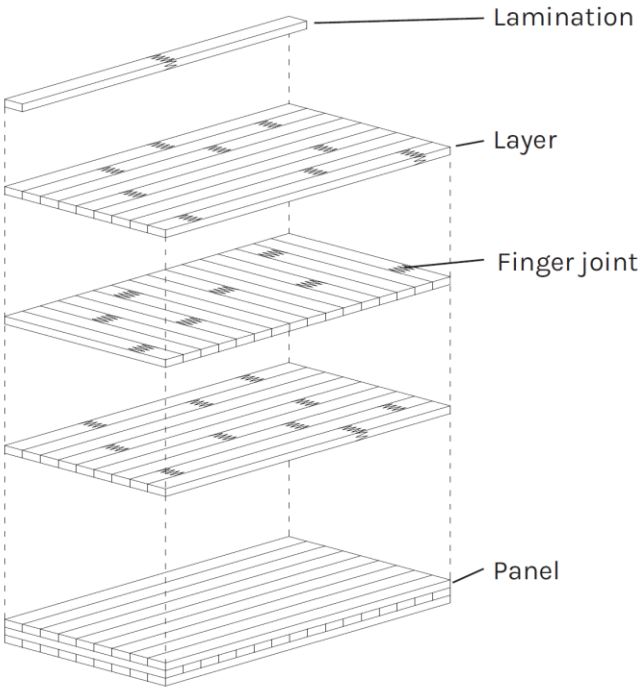


Figure 2.2: Schematic view of CLT layered configuration, reprinted from Younis and Dodoo [29]

Regarding Glulam, this material is primarily utilised for load-bearing structural frameworks [22]. It is manufactured by gluing wood laminations together with the grain running parallel to the longitudinal axis (Figure 2.3), allowing for the creation of columns and beams in a vast range of shapes and sizes.

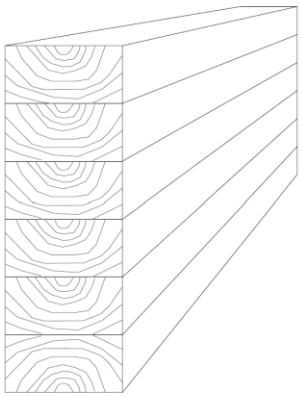


Figure 2.3: Schematic view of Glulam configuration, reprinted from Swedish Wood [22]

2.2.2. Connections

The design of connections is fundamental to the structural integrity of mass timber buildings, as these elements determine how loads are distributed through the system [30]. Beyond structural stability, the choice of connection also influences the ease of deconstruction of the building. However, the circular strategies is currently hindered by technical challenges such as the presence of coatings, adhesives and nails, that complicate the recycling process [31].

While traditional metal fasteners, such as angle brackets and hold-downs, remain the norm in the construction sector, there is an increasing shift toward innovative systems that prioritise prefabrication and rapid assembly [30]. A promising alternative to these metal elements is the use of timber-based, carpentry-type connections. By utilising wooden joints, buildings could offer a more holistic environmental benefit. Such systems are already available for wall-to-floor and panel-to-panel connections for CLT and are expected to become more competitive as they are facilitated by advanced Computer Numerical Control (CNC) machining and automated design integration.

2.2.3. CLT Buildings in LCA

Nowadays, assessing the environmental impact of buildings often involves LCA. To optimise the carbon footprint of a CLT building, different aspects can be incorporated into the LCA, including non-energy related resources, similarly to conventional buildings. Indeed, reactions resulting from industrial processes, land-use practices, combustion of fossil fuels associated with material production, transport, construction, maintenance, and deconstruction need to be considered in the analysis and optimisation of climate impacts of CLT buildings [30]. For example, optimised material use and connections in a CLT building have an impact on the carbon footprint of the production, service-life, and EoL stages. This is in line with Ahn et al. [32], who highlight the need to reduce GHG emissions across the entire life cycle. This necessitates a focus toward the EoL stage, where decommissioned structures are viewed not as a waste burden, but as a source of secondary materials that can substitute for virgin resources. The EoL stage has become a main part of LCA and its management offers an opportunity to extend the lifecycle of the building [32].

By analysing the LCAs carried out over the last 10 years on the carbon footprint of the CLT buildings, the paper of Younis and Dodoo [29] provides an overview of the feasibility of their use from an LCA perspective. The research indicates that the use of CLT instead of conventional construction material can yield average GHG emission savings of approximately 40 %, particularly when carbon sequestration and optimised EoL scenarios are considered. However, current LCAs of CLT buildings in the literature are characterised by significant variability due to inconsistent methodologies and a frequent reliance on simplified cradle-to-gate data that overlooks the complexities of the use and disposal phases. There is a critical need for an internationally recognised framework that provides a more accurate characterisation of service-life maintenance, biogenic carbon flows, and structural EoL treatments. Furthermore, the research of Jayalath et al. [33] in Australia, described in the overview of Younis and Dodoo [29], reports that although the environmental benefits of CLT buildings are limited during the building service-life compared to a conventional building, they offer significant potential for improvement during the EoL phase, through reuse and recycling.

In LCA, a key factor is the estimated lifespan of the material. In the study from Andersen et al. [34], comparing concrete and CLT building in term of environmental impacts, the lifespan of the building is considered as the same as for the concrete building. Indeed, Stora Enso, a Swedish CLT manufacture states that: "With the right building physics construction methods, there is no limit to lifetime"(Enso 2013, [34]). However, the lifespan of CLT building has not been defined yet due to a lack of research stemming from the recent development of this material. Indeed, it is known that moisture content affects the performance of timber elements by altering their mechanical

properties [30]. These conditions, combined with certain temperatures and oxygen levels, can lead to fungal degradation of the wood. This can affect the durability and lifespan of a CLT structure, as it does with other timber structures.

2.2.4. End-of-life Practices of CLT Today

As mentioned in the CLT and Glulam Section, the EoL and post-EoL stages offer an opportunity to reduce the carbon footprint of CLT structures. According to Ahn et al. [32], this opportunity is driven by several key factors that allow the preservation of the material without significant damage. These include the dimensions and configuration of panels, the types and quantity of connectors and the cost effectiveness of deconstruction over demolition. However, as said previously, environmental impact assessment of mass timber building are often based on cradle-to-gate LCAs, without taking into account building service-life and EoL stages [32]. Furthermore, the mass timber industry remains in an early stage. Most of mass timber building have not yet reached their mid-service life, so significant knowledge gaps persist regarding their long-term durability, maintenance requirements, and safety. The absence of empirical data concerning EoL trajectories and post-EoL cycle assessments presents a barrier to evaluating the circularity and environmental recovery potential of these building assemblies.

On the other hand, practices in the early building planning and design stage could lead to a higher degree of deconstruction, which could result in wider EoL options [32]. These practices include design for manufacturing, assembly and disassembly, design for deconstruction, design for re-use or recycling, and are developed in Section 2.3.

Even though mass timber circularity is still in early stages, some case studies already exist, highlighting the environmental potential and challenges of reusing CLT. For example, Passarelli [35] documented the construction of a single-storey commercial café in Japan, which was successfully built using CLT panels salvaged from a disassembled five-storey structural testing building. The project demonstrated that the reuse of CLT panels allows a significant reduction of overall GWP of the building but also revealed several logistical and design hurdles. First, working with recovered materials has required a reversed design process, adapting the building layout and dimensions to the available CLT panels rather than ordering panels to fit the design. Furthermore, because the original panels were designed with tension bolts connections, an additional remanufacturing was required to reuse the panels, resulting in almost 30 % of the wood lost as waste. The project also underscored the vulnerability of timber between the life cycles. Inadequate storage exposed the panels to moisture, causing swelling and further on-site adaptation. Ultimately, this case study illustrates that while CLT reuse is feasible and beneficial in term of environmental impacts, maximising its potential requires building system tailored for disassembly and better protocols between the life cycles.

2.2.5. Glulam Buildings in LCA

A comprehensive review of LCAs conducted over the last 10 years, by Liu et al. [36], indicates that Glulam buildings outperform steel and concrete alternatives, showing a reduction of impact in 11 to 18 impact categories. Recent research further supports that Glulam can effectively mitigate carbon emissions throughout every phase of their life cycle.

The mitigation of environmental impacts begins at the production stages [36]. As a bio-based material, the wood used in Glulam sequesters CO_2 during its growth, acting as a temporary carbon sink. The manufacturing process itself is relatively low energy demanding, requiring power for cutting, drying, and gluing, and this energy can often be sourced from electricity or biomass. While

the production of structural adhesives generates carbon emissions, the impact is minimised by the fact that adhesives constitutes only 1 % of the Glulam [22]. Although bio-based adhesives are emerging as a sustainable alternative, their adoption is currently limited due to high costs and a lack of data regarding their long-term durability [36].

During the transportation and construction phases, energy use is reduced due to Glulam high strength-to-weight ratio [36]. Indeed, Glulam is recognised as one of the most structurally efficient materials relative to its mass [22]. Additionally, the high degree of prefabrication and the use of modular assembly systems shorten the duration of on-site operations and the need for temporary facilities, thereby reducing the associated construction-site emissions [36].

During the use phase, Glulam structures provide good thermal insulation, which can lower the energy required for heating and cooling [36]. However, the environmental benefits in this phase may be partially offset by the need for maintenance such as the application of protective coatings.

Regarding the lifespan of Glulam elements, it can last up to 100 years according to Feys and Defever [37], providing a high potential for reuse.

2.2.6. End-of-life Practices of Glulam Today

Predicting the EoL remains a challenge, as most existing Glulam buildings have not yet completed their service lives, leading to modelling limitations and reliance on theoretical approximations [36]. Nevertheless, data suggests that circular strategies such as recycling and prefabrication can reduce EoL emissions by 12 to 29 % and optimise project costs. Because incineration or decomposition of wood at the EoL releases the biogenic carbon stored in it, extending the lifetime of Glulam is critical to maximising carbon sequestration.

Different EoL scenarios can be considered for Glulam: reuse, recycling, incineration with or without energy recovery, and landfilling [36]. Reuse and recycling have the most environmental benefits but despite this, the reuse of Glulam remains underdeveloped, and it is mainly incinerated or landfilled at its EoL .

To address this issue, a recent study by Feys and Defever [37] investigated the practical reclamation processes and environmental impacts of reusing Glulam structures. The authors point out that while Glulam elements possess an estimated total lifespan of 100 years, they frequently face demolition after only 30 to 60 years of use. To quantify the benefits of recovering these elements, the study employed a mixed-methods approach. Feys and Defever first analysed seven successful Belgian case studies and interviewed industry experts to map the practical steps of timber reclamation. Based on these qualitative insights, they conducted an LCA, comparing newly produced Glulam beams against four distinct reuse scenarios, ranging from direct site-to-site reuse to extensive external reprocessing. The study identified that widespread reuse is currently hindered by a lack of legal frameworks to guarantee mechanical properties, as well as severe logistical challenges regarding transport, storage, and timing mismatches between demolition and new construction. However, when these practical barriers are overcome, the quantitative results from their LCA demonstrate substantial benefits. Indeed, their study concluded that the integration of reused Glulam beams yields a 72 % to 92 % reduction in overall environmental impact compared to manufacturing new virgin beams across the full life cycle.

2.3 Design Strategies for Circularity

While utilising timber in construction can help mitigate climate change and resource depletion, the rising demand for wood can created unintended ecological pressures [38]. Specifically, intensive harvesting may compromise forest biodiversity, disrupt ecosystem services, and diminish the carbon sequestration capacity of forests. To ensure long-term sustainability, it is essential to integrate wood-based construction with CE principles.

The different strategies for circularity can be separated into two categories: upstream and downstream strategies (Figure 2.4) [39]. Upstream strategies are developed at the end of life of a building, reusing reclaimed components, while downstream strategies are beginning during the design phase, thinking about how materials will be meant to be reused.

Concerning downstream strategies, two different approaches emerge: Design for Disassembly (DfD) and Design for Adaptability (DfA).

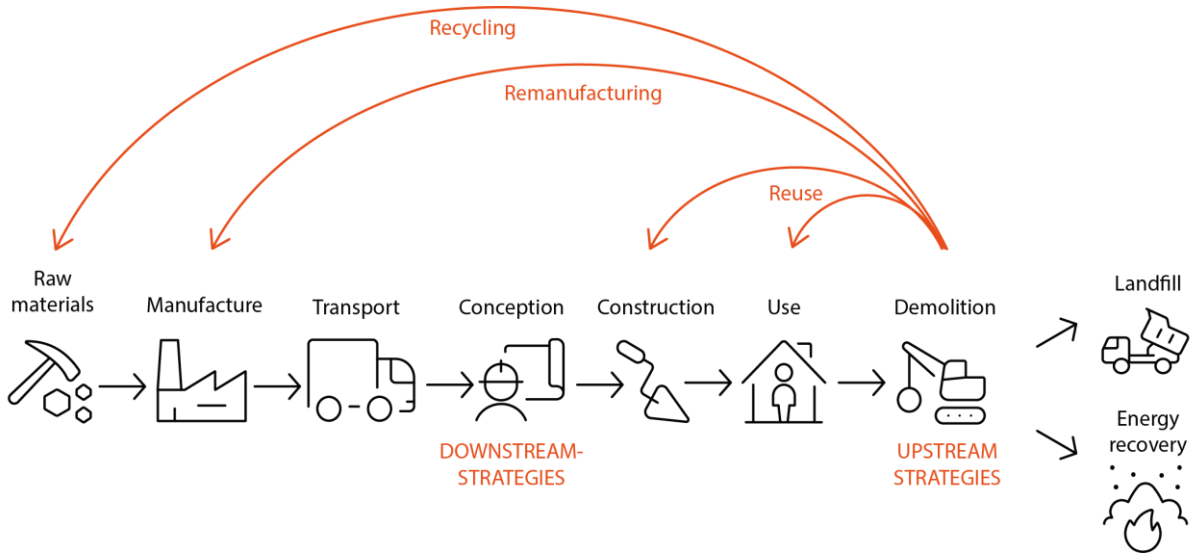


Figure 2.4: Linear and circular construction (illustration done by the authors)

2.3.1. Design for Disassembly (DfD)

DfD is a sustainable approach that prioritises the ease of deconstruction [31], [40]. By planning for the separation of a building parts from the design phase, this method ensures that materials can be effectively reclaimed for recycling or reuse at the end of the building life cycle, increasing circular economy in the built environment.

Tsoka and Tsikaloudaki [40] summarised a serie of guidelines for an efficient DfD. These include having a detailed documentation about the materials and the way of disassembling them to improve the effectiveness of the dismantling process. The choice of materials should take into consideration their future impacts and their potential reuse or recycling value. The building layout should also be simple, with standardised dimensions and a limited number of different types of components, allowing an easier deconstruction process. The structure and the other components of the buildings should be separated, facilitating the disassembly of one layer without affecting others. Finally, the type and number of connections is also important. Mechanical connectors should be prioritised to promote disassembly, and they should be easily accessible and visible to allow workers to understand easily how to disassemble the building. The fewer they are, the more rapid the deconstruction will be too.

2.3.2. Design for Adaptability (DfA)

DfA prioritises the building adaptability. "Adaptability" has a lot of different definitions, but the main principle reported by this term is the building ability to adapt to changes that will occur over its lifetime [40]. DfA should take into consideration different EoL scenarios from the earliest stages of the building design process and allow easy modifications at any stage of the building lifecycle, thereby promoting a longer lifespan with minimal component replacement requirements.

Different measures that encourage adaptability have been highlighted by Tsoka and Tsikaloudaki [40] . First, layering the components of the building and making them accessible allows replacement or modification of one layer without compromising others. A reserve capacity, both for structural elements and services spaces, can also help adaptability, allowing the building to accommodate future higher loads or different services. Having accurate information about the materials, appropriate materials, simple building layout and mechanical connections is also very important, as for DfD, to facilitate adaptation projects.

The two strategies can be implemented together to improve the implementation of circular economy in the built environment.

2.3.3. DfD and DfA in Timber Construction

Concerning wood products specifically, Piccardo and Hughes [38] identified strategies to increase the reuse of wood materials in buildings. Their study aims to explore how architectural practices currently incorporate the recirculation of wood products by identifying specific design patterns for facilitating the future reuse of materials and for utilising already salvaged wood. A qualitative methodology was used, conducting semi-structured interviews with designers of nine case-study buildings to capture their practical experiences and challenges. The findings reveal three main design patterns for future reuse: joint reversibility, separation of parts, and prefabrication. Concerning utilisation of reclaimed wood, two patterns were identified: salvaged wood recovery and adaptable building layouts. However, the results also highlight challenges such as increased design costs, uncertainties regarding the structural integrity of salvaged wood and difficulties in meeting modern energy and acoustic standards. This study confirms the previous findings regarding circular strategies, whilst applying them specifically to timber. However, their qualitative, case-study-based method captured valuable designers insights, but it relies on a small sample, without having any quantitative data, emphasising the lack of rigorous metrics to properly measure the actual benefits of these building strategies.

2.4 Life Cycle Assessment (LCA)

Evaluating the sustainability of a building requires a holistic analysis of its economic, social, and environmental impacts from its inception to its eventual removal [29]. LCA is the standard methodology for quantifying these environmental effects, structured according to the ISO 14040 [41] framework. A building life cycle encompasses every phase of its existence, starting with the extraction and processing of raw materials, moving through construction and its functional service life, and concluding with demolition and waste management. According to the EN 15978 [42] standard, these activities are categorised into four distinct stages (Table 2.1):

- Module A1-A5: raw material production, transportation and the construction process.
- Module B1-B7: operational service life, including maintenance.
- Module C1-C4: EoL stage, such as demolition and disposal.
- Module D: potential environmental benefits or loads resulting from the reuse or recycling of materials beyond the initial building boundaries.

Table 2.1: The different stages of the life cycle, according to EN 15978 [42]

Product stage			Construction stage		Use stage								End of life stage				Consequences beyond the system boundary
A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	B8	C1	C2	C3	C4	D
Raw materials	Transport of raw materials	Production	Transport to construction site	Construction, building and assembly work	Use	Maintenance	Repair	Replacement	Conversion	Energy use during operation	Water consumption during operation	Transport during operation	Demolition	Transport	Waste treatment	Disposal	Material, energy recovery, reuse of materials and export of self-produced energy

However, it has been observed that the life cycle modules included in an assessment often vary, as they are highly contingent upon the study specific goal and scope.

2.4.1. Accounting for Reuse

To accurately evaluate whether CE strategies really deliver on their goal of environmental improvement, Haupt and Zschokke [43] emphasise the necessity of LCA. Without this quantitative measurement, CE initiatives risk to inadvertently worsen environmental impacts rather than improve them. However, integrating CE principles in LCA presents significant methodological challenges [44]. Because traditional LCA is designed to assess the primary function of a single product system, linearly, it struggles to account for the multi-functional nature of CE practices like recycling and reuse. Consequently, to effectively evaluate circular economy, LCA methodologies must transition from a single life-cycle perspective to a multiple life-cycle approach that accommodates continuous material loops. A major complication in this shift is determining how to fairly allocate the environmental burdens and benefits when materials are shared across several interconnected product systems.

While several different methods exist, there is a lack of unified method to credit reuse in LCA. De Wolf et al. [39] compared different methodologies to calculate the environmental impact, specifically the GWP, of reusing building elements. Six different allocation methods were applied to a case study, the Kopfbau Halle 118 building in Switzerland, constructed almost entirely with reused materials, to demonstrate the discrepancies between the methods [39]. The assessments are modelled across a component first life cycle, an intermediate life cycle and its final life cycle. The different assessment methods studied in the paper are the following:

- Cut-off method (100:0): It allocates 100 % of the production impacts to the first life cycle and 100 % of the final disposal impacts to the last life cycle. It rewards the use of already used materials (upstream reuse) but not the buildings that are designed for future reuse.

- End-of-life method (0:100): It attributes all production and end of life impacts to the last life cycle. It encourages designing for future reuse but heavily relies on the assumption that components will be reused in unpredictable future scenarios.

- Distributed allocation: It evenly distributes the production and EoL impacts across all the life cycles of the component. It is highly dependent on the number of reuse cycle, which is difficult to predict.

- European Commission Environmental Footprint: It splits the production and EoL impacts in a 50:50 ratio, shared between the first and last life cycle, mitigating the uncertainty of predicting the number of reuse cycle.

- Degressive: It is a hybrid approach blending the two previous methods, allocating impacts degressively over intermediate reuse stages.

- SIA 2032: It is a Swiss norm that divides production and EoL impacts based on time. It calculates a ratio of the component actual lifespan in its current building against its expected total lifespan.

De Wolf et al. [39] found that applying different allocation methods to the same building resulted in GWP values that can be radically different. Depending on the chosen methodology, the calculated GWP per gross floor area varied by a ratio of one to two for the first life cycle, and one to three for the last life cycle. The study concludes that the current LCA quantification methods do not provide consistent results, which hinders objective comparison between different circular strategies. Furthermore, the findings highlight that the current scope of LCA is too limited and do not take into consideration key characteristics specific to reuse.

In another study, conducted by Eberhardt et al. [44], the distributed allocation was used for all reused materials in a case study of a Danish office building designed for disassembly. The findings indicate that the environmental impacts savings from DfD vary across impact categories and depends on the building materials, component service life, and the number of future reuse cycles.

2.4.2. Accounting for Biogenic Carbon

Taking into consideration the biogenic carbon is a controversial issue, with various methodologies existing [45]. While sustainable forestry is often labelled as carbon neutral, this assumption is increasingly challenged. As trees grow, they sequestrate carbon, i.e. remove CO_2 from the atmosphere, and act as temporary storage by maintaining the sequestered carbon for a specific duration [46]. This storage helps mitigate climate change by delaying the presence of CO_2 in the atmosphere, though this carbon is eventually re-emitted at the end of the product life.

Unlike fossil fuels, which represent a one-way release of carbon, managing biospheric carbon involves a dynamic, two-way exchange where carbon is both absorbed and emitted. Even if the total amount of carbon absorbed equals the amount released, the timing is critical. Whether sequestration precedes emission (such as planting a forest before harvesting) or emission precedes sequestration (such as burning wood and waiting for regrowth), the sequence determines the actual concentration of CO_2 in the atmosphere. Because climate impacts depend on how long GHG reside in the air, these different timing trajectories result in different levels of cumulative radiative forcing, meaning they trap different amounts of heat over time. Furthermore, the true atmospheric concentration at any given moment is not just a simple calculation of human emissions minus tree growth; it is constantly influenced by how CO_2 interacts with the Earth natural carbon reservoirs, like oceans and soil.

Despite these complexities, many LCA models for the wood products cascade, i.e. multiple material utilisation of wood products before their conversion into energy, either ignore biogenic carbon entirely or simplify the process by assuming that growth and emissions cancel each other out, thereby neglecting the impact of timing [45].

Garcia et al. [45] identified several methods to consider biogenic carbon and compared them, in combination with different EoL allocation approaches. The two most simple accounting methods for biogenic carbon are the Zero-GWP, which simply does not take into account biogenic carbon, and the Fixed GWP, which assigns the same characterisation factor regardless of time [45]. Although the physical life cycle of a wood product involves a clear time delay between absorbing carbon during forest growth and emitting it at disposal, the Fixed GWP method ignores this intermediate storage phase. It actively counts both the initial carbon uptake and the final emission, but because it gives no mathematical credit for the years the carbon is temporarily stored in the product, it treats the two events as happening simultaneously, meaning they always perfectly cancel each other out to a net-zero impact.

Other methods, presented in the study of Garcia et al., such as ILCD or PAS 2050 give credit for each year during which carbon was stored in wood, with a starting boundary when the tree is harvested. More complex methods exist too such as GWPbio, which evaluates the timing of carbon storage in both the manufactured product and the natural forest. The most flexible and comprehensive method is the dynamic LCA, which tracks emissions and absorptions of carbon dynamically over time, allowing the user to account for the timing of carbon storage in both the product lifetime and the forest growth.

2.5 Research Gap

Despite the established environmental benefits of timber construction and the theoretical development of CE principles, several critical research gaps remain regarding the quantitative evaluation of these strategies in practice.

First, while the literature extensively covers the production phase of CLT and Glulam, current LCAs are predominantly simplified cradle-to-gate studies that frequently overlook the complexities of the use and disposal phases. Given the recent emergence of mass timber construction, there is a lack of data concerning the EoL trajectories and post-EoL stages of mass timber, which creates a barrier to fully understanding the environmental recovery potential of these buildings.

Second, although design strategies to improve timber circularity have been clearly identified by architectural practitioners, there is a lack of quantitative metrics to measure their actual climate benefits. Without quantitative LCA measurements evaluating continuous material loops, circular initiatives cannot be objectively validated and risk inadvertently worsening environmental impacts.

Finally, significant methodological inconsistencies persist regarding how to accurately assess these material loops. Current LCA frameworks struggle to fairly allocate environmental burdens across interconnected product systems, and many models rely on simplified, time-blind biogenic carbon accounting methods that fail to capture and mathematically reward the delayed emissions achieved through prolonged material storage in the anthroposphere.

This thesis directly contributes to address these gaps by quantitatively evaluating the specific GWP benefits of DfD strategies for multi-storey timber buildings. By modelling CLT panels across three distinct life cycles and comparing varying EoL allocation methods, this research provides data on the climate mitigation potential of improving the circularity of the construction sector.

3 Case Study Description

Moving from the theoretical framework to practical application, this section describes the case study project.

HasleTre is a 3000-square-meter wooden commercial building (Figure 3.1) located at Haslevangen 18, on the edge of central Oslo. Developed by Höegh Eigendom with Oslotre acting as the architects, the project was initiated in 2019 and completed in 2022 [47]. Spanning five floors, which include a basement and roof terraces, it is recognised as Norway and world first timber office building explicitly designed and constructed for future disassembly and reuse. The project, heated by renewable district heating, achieves high environmental ambitions, reducing the GHG emissions from material by 47 % (BREEAM) and from materials and energy demand by 50 % (FutureBuilt ZERO) [48].



Figure 3.1: Picture of HasleTre, taken by Dmitry Tkachenko, with permission of Oslotre [47]

The building relies on an optimised primary structural system (Figure 3.2 and Figure 3.3) consisting of CLT slabs, a Glulam post-and-beam framework and finished with a shingle timber facade [47]. The interior spaces are organised around a central core and subdivided by a 5x5-meter structural grid. This grid provides, according to the architects, flexibility for reconfiguring floor layouts without requiring exterior or structural modification, with the aim of extending the building lifespan, independently of the tenants and their specific needs [49].

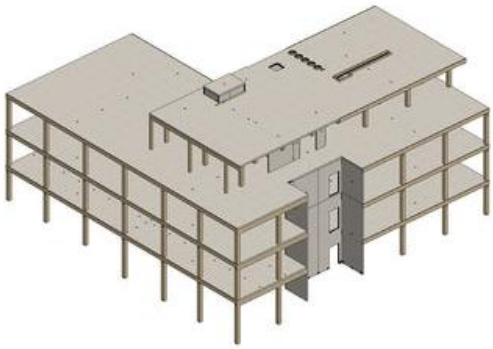


Figure 3.2: Illustration of the structure of HasleTre, with permission of Oslotre [47]



Figure 3.3: Picture of the construction of HasleTre: the structural elements, with permission of Oslotre [47]

The CLT floor slabs utilise standard 3x15-meter master elements, which is done to minimise production cut-offs and to maintain standard dimensions for future buyers [49]. Furthermore, to accommodate technical services like ventilation and sprinkler systems without compromising the future utility of the wood, the Glulam beams are not perforated. Instead, the inherent strength of the CLT slabs is utilised to span over the beam structure, leaving dedicated space for open technical infrastructure, as illustrated on Figure 3.4.

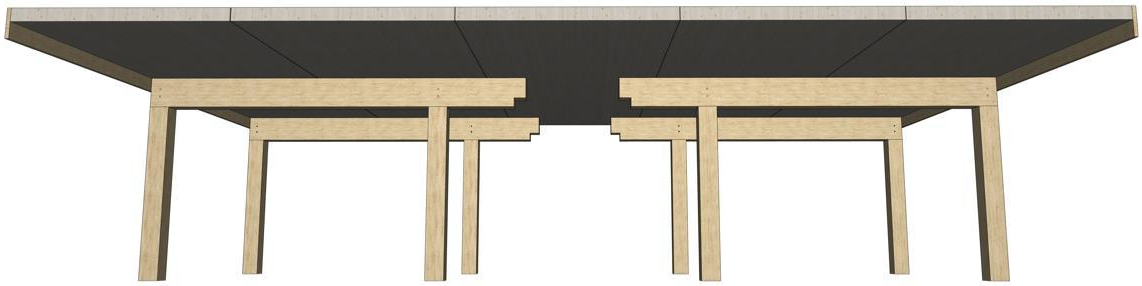


Figure 3.4: Illustration of the non-perforated Glulam beams of HasleTre, with permission of Oslotre [47]

This area is fitted with a Granab subfloor system, an adjustable dry-flooring solution that uses steel battens and acoustic dampers to create an installation cavity, as shown in Figure 3.5 [50]. By using standard manufacturing formats and dimensions and preserving their structural integrity, the timber structure is designed to preserve the value of the components for second-hand use.



Figure 3.5: Illustration of the Granab subfloor system used in HasleTre, with permission of Granab [50]

The preservation of the wood components is also achieved by the use of timber-to-timber joints to replace traditional metal connections [49]. It allows the wood element to be reworked and remodelled with industrial tools at the end of the building life, avoiding the danger of embedded steel in the component. To this end, three types of all-wood joints were used:

- Beech dowels to lock the post-and-beam nodes that can be drilled out during disassembly (Figure 3.7).
- X-fix connectors to join the CLT slabs in the walls and floors (Figure 3.6 and Figure 3.7). These reversible plywood dovetail connectors replace traditional metal screws and accelerate the on-site assembly process [51]. Installation is performed by hammering the connectors into pre-milled slots to create a dry and mechanical wedge joint. This approach aims to preserve CLT panels integrity.
- Beech nails for fixing the internal ash panels and the external pine cladding [52].



Figure 3.6: Picture of a X-fix connector in HasleTre project, taken by Einar Aslaksen, with permission of Oslotre [47]

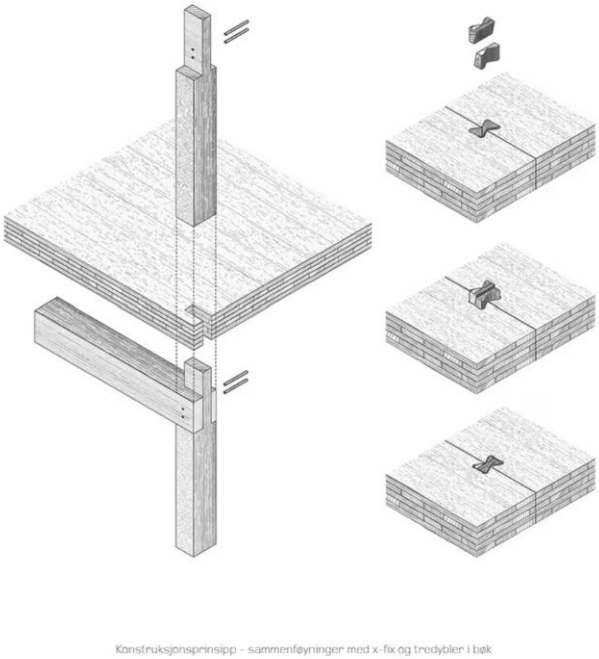


Figure 3.7: Illustration of the wooden connections of HasleTre (left: wooden dowels, right: X-fix), with permission of Oslotre [47]

These design strategies allow, according to one of the architects of the project, to have 100 % of the CLT and Glulam that could, theoretically, be recovered and reused.

The circular design framework extends to the building technical systems and interior fit-out, incorporating salvaged components for ventilation units, acoustic ceiling panels, sanitary installations, and second flooring. This strategy is further reflected in the furniture selection, which consists of 60 % upcycled pieces and 40 % biodegradable items manufactured from industrial waste streams [53]. Furthermore, the architects provided a disassembly manual of the building supporting the future disassembly of HasleTre.

To conclude, by adhering to standard component dimensions, preventing beam perforations, and utilising wood connections, HasleTre tries to preserve the structural integrity and second-hand financial value of its materials for future reuse.

4 Materials and Methods

To operationalise the research framework established in the previous sections, this chapter is structured into four sequential phases. At first, a FGD is conducted with industry experts to gain insights on different circular solutions for timber buildings. Then, a comparative LCA is done to compare the HasleTre building with its linear equivalent, without modular elements and wooden reversible connections, to evaluate the advantages of these solutions on the environmental impacts. After that, the CLT panels are isolated, and their environmental impact is assessed through three successive life cycles. Finally, a sensitivity analysis is conducted to test the reliability of the LCA and address the uncertainties of future scenarios.

4.1 Focus Group Discussion

As part of a broader research project investigating circular wood construction, a FGD is conducted to provide qualitative data to this thesis. The primary objective of the session was to evaluate and rank technical solutions designed to improve the circularity of multi-storey timber buildings in Scandinavia.

Held on March 5, 2026, the FGD involved five wood industry experts representing various stages of the construction chain, including an architect, a structural engineer, technical representatives, and a contractor. The discussion was organised into three technical pillars: service life extension, deconstruction/recoverability, and reusability. During the meeting, participants discussed and evaluated specific examples, such as reversible screws and modular elements, providing insights based on the structural behaviour, technical, and economic feasibility. Guided by a semi-structured format, the moderator used open-ended questions to stimulate experts interaction while ensuring that all three pillars were addressed. This inclusive approach was intended to balance the various professional perspectives represented. Meanwhile, during the discussion, the authors of this thesis served as note-takers, documenting key technical insights, nuances and group dynamics; the session was not recorded.

To supplement the verbal discussion, a digital Wooclap [23] was used to rank the participants views on the most effective technical solutions and the biggest industry obstacles. Results were analysed using a weighted ranking method where points were assigned to each element based on its position in an individual participants list (with the top-ranked item receiving the highest score). These points were then summed to establish the final group hierarchy. This mixed-methods approach allowed the research to capture both nuanced qualitative insights and quantify priorities. These expert insights acted as a filter, helping to narrow the thesis focus toward the most impactful circular strategies available in the Swedish and Norwegian market.

While this session followed the core principles of a FGD, relying on group interaction to generate data, it is important to note certain departures from the conventional methodology. As only one session was conducted in this case, the results are treated as preliminary and are understood to reflect the unique dynamics of this specific group. Furthermore, it must be acknowledged that the participants were already advocates for circularity. This created a pre-existing alignment with the environmental value of the discussed strategies, which probably influenced the group dynamics. The findings were not intended to stand as a definitive consensus but were instead used as a scoping tool to refine the research questions and validate the assumptions used in the quantitative LCA.

4.2 Scope

The selection of a case study, in this instance the HasleTre office building in Oslo, as described in Section 3, allows this research to be anchored in a real-world context. As the mass timber industry is still relatively young, many buildings have not yet reached mid-service life, leaving open questions about their long-term performance and EoL impacts [32]. A case study approach is therefore necessary to bridge the gap between theoretical circularity and actual construction practices, providing a foundation for exploring the recovery potential of structural timber elements.

HasleTre was chosen specifically because it is recognised as a pioneering project, the world first timber office building explicitly designed for future disassembly and reuse, while being firmly anchored in the Scandinavian context. It serves as a best practice model by integrating specific technical solutions, such as reversible connectors and all-wood joints, which replace traditional metal fasteners. The existing building allows the study to rely on a combination of technical documentation, architect data, and insights from industry experts gathered during the FGD.

4.3 Comparative LCA

The second phase of the method employs a comparative LCA to evaluate the environmental benefits of the circular design of the case study. This stage translates the qualitative insights from the FGD, and the data collected from HasleTre into a technical comparison between the circular case study and the linear reference model. The following subsections define the scope, system boundaries and the specific modelling method used to calculate their respective GWP.

4.3.1. Scope, Boundary and Data

The comparative LCA aims to quantify the benefits, in terms of GWP, of using reversible wood connections (X-fix and wood dowels) and modular elements. To this end, the HasleTre office building (3000 m²), designed using DfD principles with reversible connections, is compared with a theoretical equivalent, without these circular solutions and using conventional metal fasteners. This comparison specifically takes into account the higher reusability rate, i.e. capacity of the material to be reused at the end of life of the building, resulting from the modularity and reversibility of the assemblies.

The analysis is carried out at the scale of the entire building, with a functional unit of 1 m² of total floor area for a calculation period of 50 years.

The system boundary is detailed in Table 4.1. It intentionally excludes operational energy (B6) and water use (B7) to isolate material-related impacts. Furthermore, B1 to B3 are excluded due to current methodological limitations. Indeed, there is a lack of empirical data for these stages, therefore, only major replacements and refurbishment are included. The deconstruction and demolition stage (C1) are also excluded due to a lack of reliable data for this phase for CLT buildings.

Table 4.1: Life cycle stages considered in the comparative LCA (stages with a "X" are considered)

Product stage			Construction stage		Use stage								End of life stage				Consequences beyond the system boundary
A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	B8	C1	C2	C3	C4	D
Raw materials	Transport of raw materials	Production	Transport to construction site	Construction, building, and assembly work	Use	Maintenance	Repair	Replacement	Refurbishment	Energy use during operation	Water consumption during operation	Transport during operation	Demolition	Transport	Waste treatment	Disposal	Material, energy recovery ,reuse of materials, and export of self-produced energy
X	X	X	X	X				X	X					X	X	X	X

The life cycle inventory is based on data provided by the architects, including the weight of materials and their respective reusability rates. Where the exact type of a material was not specified, identical assumptions were applied to both variants to ensure consistency in the comparison. The materials were also chosen to be from Norway when possible. It includes the primary structure (foundations, beams, columns and CLT) and the building envelope (walls, roofs, slabs, partition systems, windows), as detailed in Appendix A. Furniture and service systems are not included in the analysis.

Impact calculations were performed using the One Click LCA calculation program, version 0.50.0, database version 7.6, based on the EN15804+A1 standard. Despite the use of this standard, where data is mandatory only for cradle-to-gate, EoL impacts were included in the analysis by applying industry-standard deconstruction and disposal scenarios provided by the One Click LCA database when it was not given by the selected Environmental Product Declaration (EPD). This allows for a complete assessment closing the material loop.

Finally, biogenic carbon flows associated with wood-based materials are excluded from this analysis. This exclusion is based on the assumption that the timber originates from sustainably managed forests where carbon sequestration from new growth effectively balances the carbon removed during harvesting. This decision is further justified by the complexity of its quantification, which depends on temporal carbon fluxes and the absence of a unified method, as detailed in Section 2 [30]. A dynamic LCA, necessary for a comprehensive view of these fluxes, would have gone beyond the scope of this research.

4.3.2. Life Cycle Analysis

This subsection details the life stages included in the LCA and the associated assumptions.

- Product and Construction Stages (A1–A5)

The impacts of the production stages (A1–A3) are calculated based on the project material quantities and the EPDs selected in the One Click LCA software.

For the transport stage (A4), the assumptions are based on the default "Nordic Transport" scenarios provided by One Click LCA. Details of the various scenarios applied according to the origin of the materials are available in Appendix C.

Regarding the construction stage (A5), the rates of construction site losses and waste are in line with the reference values from One Click LCA, as summarised in Appendix C. The construction site operations are calculated based on the gross floor area and using the generic data of One Click LCA "Average energy and fuel use in construction site - Nordics (per GFA), 2015". This set of data assumes an electricity use of 43 kWh/m^2 (GFA) with an emission factor of 0.034 kg CO_2e/kWh and a total use of diesel of 5.2 L/m^2 (GFA) with an emission factor of 3.24 kgCO_2e/L.

- Use Stage: Replacement and Refurbishment (B4–B5)

When component replacement is necessary, the transport scenario applied is identical to that of stage A4. The reference service lives are taken directly from the respective EPDs and are listed in Appendix C.

- End of Life Stage (C2–C4)

EoL scenarios vary depending on the nature of the materials and are detailed in Appendix C. Particular attention is paid to the reusability rates of Glulam and CLT, which differ between the two buildings to reflect DfD strategies.

- Benefits and Loads Beyond the Building Life Cycle (Module D)

Module D accounts for the net environmental benefits associated with the recovery of materials at end of life.

The energy recovered from the incineration of wood or other materials is considered as a fuel substitution for district heating.

The benefits associated with the recycling and the reuse of materials are also incorporated, enabling the quantification of the avoidance of virgin raw material extraction.

4.3.3. Model of the HasleTre Building (HT Circular)

The modelling of the HasleTre building, hereafter referred to as HasleTre Circular (HT Circular), is based on the detailed list of materials with material quantities by weight (Appendix A), provided by the architects, incorporating the specified reusability rates.

For CLT and Glulam, this rate approaches 100 %, representing the theoretical maximum recoverability enabled by the building design. This high recovery potential is justified by several strategic design choices.

As detailed in Section 3, a primary factor is the use of large modular elements and standardised dimensions, which minimise the need for resizing in future projects. Furthermore, the implementation of reversible wooden connections, such as X-fix connectors and wooden dowels

allows for non-destructive disassembly. Unlike traditional metal fasteners, these wooden joints avoid embedding steel within the elements, preserving their structural integrity and value for reuse.

While a 100 % reusability rate is adopted to evaluate the building, it is acknowledged as a theoretical upper limit. In practice, reuse may be influenced by future uncertainties such as localised moisture damages, minor damage during deconstruction, or specific adaptations required by new building premises.

In the absence of a specific EPD for X-fix connectors and beech dowels, their impact was modelled by converting their geometry into an equivalent volume or weight of birch plywood and beech, as calculated in Appendix B.

4.3.4. Model of the HasleTre Reference Building (HT Reference)

For the reference building (referred to as HasleTre Reference or HT Reference), the design is identical to that of HasleTre, except for the connections that are no longer reversible and the CLT and Glulam components that are not modular. These technical changes result in a lower reusability rate of 70 %. This figure is derived from the research of Passarelli [35], who documented a case study in Japan, where a commercial café was constructed using salvaged CLT panels from a disassembled five-storey structural testing building. In that project, 30 % of CLT panels intended for reuse were lost during the remanufacturing stages, particularly due to damage caused by traditional fasteners.

It must be acknowledged that this 70 % reusability rate represents a theoretical upper limit for standard construction. In the Passarelli [35] study, the source building was a simplified structure without finishing materials or technical installations. In a standard commercial or residential building, materials such as adhesives, coatings, or integrated HVAC systems typically contaminate the timber or complicate disassembly. Consequently, while this study assumes 70 % recovery for the reference building to establish a safe comparative baseline, the actual reuse rate in conventional industry practice is likely to be significantly lower.

In this comparative scenario, reversible connections are replaced by conventional metal fasteners, as detailed in Appendix B.

4.4 Multicycle Evaluation of CLT Panels

The multi-cycle assessment evaluation focuses on CLT panels, over three consecutive use cycles, as they represent the building primary structural material and the core of its DfD strategy. Because of a single LCA lack of capturing the long-term impact of a dismantlable structure, this multi-cycle simulation is necessary to investigate initial and final emissions can be shared across consecutive life cycles.

The One Click LCA simulation isolates the CLT structural panels from the whole building Comparative LCA conducted in Section 4.3. The evaluation is conducted following the three most used allocation method (100:0, 0:100 and distributed) based on De Wolf et al. [39]. Their specific methodology is explained here after, in Sections 4.4.3 to 4.4.5.

4.4.1. Scope and Boundary

The functional unit is 1 m^3 of CLT maintained over a total period of 150 years. The three consecutive life cycle have a 50-year lifespan, and it is assumed that 100 % of the CLT mass is recovered and reused in each iteration. While useful for theoretical modelling, this approach departs from reality, where CLT panels would not typically all be reused in the same building across

successive building lifecycles.The system boundary is the same as the previous LCA, detailed in Table 4.1. The results will be given in term of GWP.

The life cycle inventory takes into account 1154 m^3 of CLT, with a reusability rate of 100 %, as taken from the material list. The same standard (EN15804+A1) is used as previously, and biogenic carbon is still not considered for the same reason as in the previous LCA.

4.4.2. Life Cycle Analysis

The definition of what is taken into account in each stage is slightly different from Section 4.3.2. covering Comparative LCA.

- Product and Construction Stages (A1–A5)

The impacts of the production stages (A1–A3) are calculated based on the quantity of CLT (1154 m^3) with the EPD chosen in One Click LCA being "Cross-laminated timber (CLT), 460 kg/m^3, Krysslimt tre (Splitkon)".

For the transport stage (A4), the assumptions are based on the default "Nordic Transport" scenarios provided by One Click LCA, for the CLT this represent 130 km by trailer, as detailed in Appendix C.

For the construction stage (A5), only the waste on site is considered due to a lack of data on the environmental impact of construction for CLT only. The rate of wastage is kept with the default value of One Click LCA, being 10 %.

- Use Stage: Replacement and Refurbishment (B4–B5)

For CLT, no replacement is needed, these stages equal to zero and are thus not considered in the results.

- End of Life Stage (C2–C4)

EoL scenario modelled in One Click LCA is wood incineration, as it is what is going to happen at the end of the three uses of CLT. Reuse will be considered with the different allocation methods.

- Benefits and Loads Beyond the Building Life Cycle (Module D)

In this case, Module D accounts for the loads of reuse, disassembly and remounting as well as transport between the deconstruction site and the new construction stage. To calculate this, the environmental impacts of stage A5 is doubled and those of stage C2 are added, as done by De Wolf et al. [39].

The benefits beyond the building life cycle are thus not taken into account in module D, both for wood incineration and reuse.

4.4.3. Cut-off Method (100:0)

The cut-off method attributes 100 % of the production impacts to the first life cycle, and is consequently more advantageous to assess building that are reusing materials rather than designing building for future reuse [39].

The total environmental impact is calculated as follows in Equation 4.1:

$$I = (1 - R_1)I_P + I_C + (1 - R_2)I_D + R_1 I_R \qquad \textit{Equation 4.1}$$

Where I_P is the environmental impact of production (A1-A3), I_C is the environmental impact of construction (A4-A5), I_D is the environmental impact of disposal (C2-C4), I_R is the environmental impact of reuse, R_1 is the part of reused materials in the building and R_2 is the share of recovered materials at the end of the life cycle.

As explained in the section above, the impact of reuse can be calculated as follows (Equation 4.2):

$$I_R = I_{C2} + 2\, I_{A5} \qquad \textit{Equation 4.2}$$

Where I_{C2} is the environmental impact of stage C2 and I_{A5} is the environmental impact of stage A5.

As the reuse rate is considered as 100 % between the use cycles, for the first use cycle, the share of reused material is 0 and the share of material reused at the end of life is 100 %. For the second use cycle, 100 % of CLT is reused and will be reused again. For the last use cycle, 100 % of the CLT is reused and none of it will be reused at the end of its life. These values are summarised in Table 4.2.

Table 4.2: Values of R_1 and R_2 for each use cycle

	R_1	R_2
First cycle	0	1
Intermediate cycle	1	1
Last cycle	1	0

To be able to understand how the impacts are distributed, they are broken down into the different life cycle stages, as detailed in Table 4.3.

Table 4.3: Values for each life cycle stage and use cycle for the cut-off method

	A1-A3 $(1 - R_1)I_P$	**A4-A5** I_C	**C2-C4** $(1 - R_2)I_D$	**D** $R_1 I_R$
First cycle	I_P	I_C	0	0
Intermediate cycle	0	I_C	0	I_R
Last cycle	0	I_C	I_D	I_R

4.4.4. End-of-life Method (0:100)

The End-of-Life method attributes 100 % of the production and disposal impacts to the last use cycle, which gives more benefits to building designing for future reuse [39].

The total environmental impact is calculated as follows (Equation 4.3):

$$I = (1 - R_2)I_P + I_C + (1 - R_2)I_D + R_2 I_R \qquad \textit{Equation 4.3}$$

Where the value of R_2 are the same as for the cut-off method, described in Table 4.2.

The value for each life cycle stage and use cycle is given in Table 4.4.

Table 4.4: Values for each life cycle stage and use cycle for the End-of-Life method

	A1-A3 $(1 - R_2)I_P$	**A4-A5** I_C	**C2-C4** $(1 - R_2)I_D$	**D** $R_2 I_R$
First cycle	0	I_C	0	I_R
Intermediate cycle	0	I_C	0	I_R
Last cycle	I_P	I_C	I_D	0

4.4.5. Distributed Method

The distributed method allocates an equal amount of production and EoL impacts in all life cycle [39]. It is thus a function of the number of use cycles, which can be hard to define and impacts the reliability of the results.

The total environmental impact is calculated as follows (Equation 4.4):

$$I = \frac{I_P}{n} + I_C + \frac{I_D}{n} + I_R \qquad \textit{Equation 4.4}$$

Where n is the number of use cycles.

The values for each stage and each use cycle are detailed in Table 4.5.

Table 4.5: Values for each life cycle stage and use cycle for the distributed method

	A1-A3 $\frac{I_P}{n}$	**A4-A5** I_C	**C2-C4** $\frac{I_D}{n}$	**D** I_R
First cycle	$\frac{I_P}{3}$	I_C	$\frac{I_D}{3}$	I_R
Intermediate cycle	$\frac{I_P}{3}$	I_C	$\frac{I_D}{3}$	I_R
Last cycle	$\frac{I_P}{3}$	I_C	$\frac{I_D}{3}$	I_R

4.5 Sensitivity Analysis

The final phase of the method involves a sensitivity analysis, designed to assess how variations in the building lifespan and EoL scenarios influence the overall GWP results of the comparative LCA (Section 5.2).

4.5.1. Lifespan

To evaluate the environmental efficiency of building longevity versus material reuse, a sensitivity analysis is conducted comparing two different temporal scenarios. This analysis aims to determine if extending the service life of a structure is more effective than deconstructing and rebuilding it, even if circular strategies are integrated in the building.

The methodology consists of a short comparative LCA of the whole building, varying its lifespan. The first scenario consists in a single building iteration of HT Circular with a 100-year service life, utilising DfD principles. The second scenario consists in two consecutive 50-year HT Circular iterations. In this case, the structural CLT panels and Glulam beams are recovered at the end of the first 50-year cycle and re-integrated into a second building of identical volume. The modelling assumes that while the primary timber structure is reused, the second 50-year iteration requires a full replacement of secondary components in Stage A1-A3.

The results for each scenario were calculated using One Click LCA, although the 50+50 scenario was created manually by combining two individual scenarios in One Click LCA. The boundary and data were the same as for the Comparative LCA, as described in Section 4.3.

4.5.2. End-of-Life Scenarios

To assess the impact of different EoL scenarios for CLT and Glulam, a sensitivity analysis is performed as part of the whole-building comparative LCA.

This analysis evaluates whether material reuse provides a more favourable module D than alternative EoL scenarios such as landfilling and informal recycling. The objective is to validate the findings from the initial comparative LCA (Section 5.2), which suggested that the higher credits observed for incineration were specifically driven by energy recovery assumptions.

Four distinct EoL scenarios were evaluated, summarised in Table 4.6. The first two scenarios, HT Circular (100 % reuse) and HT Reference (70 % reuse and 30% incineration) serve as the baseline. To further investigate the environmental impacts of the EoL of the remaining wood fraction, two additional EoL scenarios were tested:

- Wood landfilling: disposal where wood undergoes anaerobic decomposition.
- Informal recycling: recovery and downcycling of wood waste outside of formal waste management systems.

Table 4.6: Summary of the different EoL scenarios compared in the sensitivity analysis

Name	Based on	Reuse	Connections	EoL
HT Circular	HT Circular	100 %	Wooden reversible connections	Reuse as material
HT Reference	HT Reference	70 %	Metal fasteners	Wood incineration
HT Ref, Landfilling				Wood landfilling
HT Ref, informal recycling				Informal recycling

The results for each scenario were calculated using One Click LCA with the same boundary and data as the Comparative LCA, as described in Section 4.3.

5 Results

This chapter presents the results derived from the four-stage research methodology, integrating both qualitative and quantitative data to assess the environmental impact of circular timber construction. The results are structured into four primary sections, including the FGD, the comparative LCA, the multi-cycle evaluation of CLT panels, and the sensitivity analysis.

5.1 Focus Group Discussion

Based on the FGD, described in Section 4.1, the results provide an overview of the technical and systemic factors influencing circularity in multi-storey wood buildings. The group, comprising experts such as an architect, a structural engineer, technical representatives, and contractors, focused the discussion on the three pillars in the timber construction sector: deconstruction, reusability and service life extension (Table 5.1).

Table 5.1: Concepts, focus, and examples of circular technical solutions presented to FGD participants.

Concept	**Focus**	**Examples**
Deconstruction and recoverability	Facilitating the non-destructive disassembly of materials at the end-of life stage	• Reversible systems (bolts, X-fix, screws, nails) • Wood dowels and wood nails
Reusability	Enabling the reuse of structural components in their original form without reprocessing	• Modular elements and pillars • Volumetric design • Dry-assembled wall layers (avoid foam and glue)
Service Life Extension/Adaptability	Extending building utility and lifespan	• Non-load-bearing partitions • Accessible service voids (ceiling and floor systems)

Regarding the deconstruction and recoverability, while the concept of dismountable connections is recognised as important by several experts, an architect expressed reservations concerning long-term structural performance. This participant questioned how wood movement and shrinkage over years would affect the behaviour of the joints, a concern that was acknowledged by a technical representative. The X-fix connector was identified by the architect as a high-performing solution for its ability to align plates and take all shear forces without leaving metal residues, which can reduce structural integrity. Whilst the technical representative discussed reusable connections, the structural engineer showed more interest in the reuse of a building main structural element. This led to a discussion where the contractor and structural engineer agreed that prioritising demountable systems during the initial design phase is essential for successful future recovery. The contractor also pointed out that standardised dimensions could improve recoverability, but slabs are easier to standardise than walls because, in his view, there are no standard heights.

In the discussion on reusability, a technical representative argued that modular elements (slabs and pillars) offer more flexibility and architectural expression than volumetric design, i.e. prefabricated modular cubes as shown in Figure 5.1. According to the contractor, these are more suited to appartements. The architect emphasised that for building component to be truly reusable, every layer must be capable of being separated, making the avoidance of foam and glue an absolute necessity to facilitate material reuse or replacement of the most vulnerable building layers.

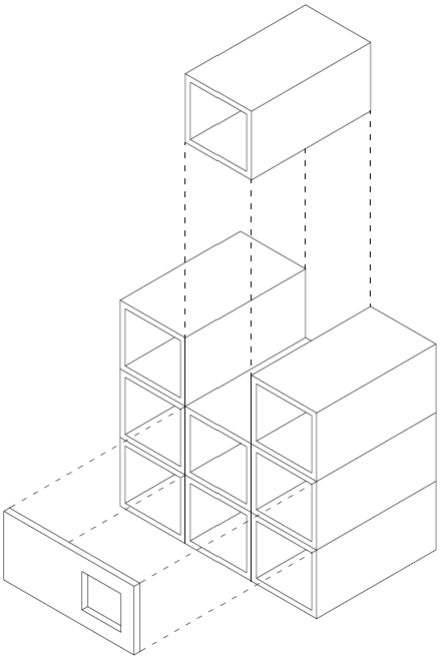


Figure 5.1: Illustration of volumetric design, reprinted from Timber Development UK [54]

The discussion regarding building longevity and the role of flexibility revealed a tension between standard industry practices and the requirements for circular economy. Indeed, the contractor questioned the conventional 50-year lifespan used for building calculations, as for him structure should be designed to last at least 100 years. However, a structural engineer noted that they are often locked by customers contracts that prioritise low-cost and optimised buildings over those designed for long-term reusability. In reaction to that, the architect emphasised the importance of built-in flexibility through spatial allowances. For example, extra space for HVAC systems can prevent a building from becoming outdated within 15 years due to changing needs. This was supported by a technical representative who observed that a shift in how people live and work now require a much higher degree of inherent flexibility in buildings. Finally, the participants agreed that while buildings have the potential for a longer lifespan, achieving this requires a shift in the design order from the client and a collaborative effort throughout the value chain.

Whilst the results presented above are the outcome of a group discussion, the following results are the outcome of the participants individual contribution to a ranking exercise presented in Appendix D, which have been compiled into an overall ranking. They prioritised technical solutions and systemic barriers presented in the Wooclap form [23], based on their industry experience. Regarding the most effective technical strategies for enhancing circularity, the group ranked modular elements as the highest priority, followed by reversible systems and the implementation of non-load-bearing partitions. When evaluating the obstacles to reusing structural materials from existing buildings, the participants identified procurement as the most significant challenge, followed by regulatory hurdles such as Boverket's Building Regulations (BBR) and CE classification, as well as the absence of established marketplaces and forums for value-chain dialogue. Looking to the challenges of making future buildings more circular, the experts concluded that unsuitable organisational and design processes represent the primary barrier, with national laws[23], regulations, and procurement remaining secondary but critical hurdles. These rankings suggest that, while technical solutions are renowned, the timber industry transition to a circular model is more restricted by procedural and regulatory framework rather than a lack of technology.

In summary, the insights gathered from this FGD serve as the qualitative foundation for following phases of this research. Transitioning from the reported expert dialogue to the specific application within this thesis, these findings act as a filter to refine the technical focus and validate the parameters of the upcoming quantitative analysis. Specifically, the expert validation of modular elements and reversible systems as top-ranked solutions encourage the focus of this study on these specific technical components, such as X-fix connectors. Furthermore, the professional feedback regarding long-term structural performance and the potential for a 100-year building lifespan provides a "reality check" for the technical assumptions and service-life parameters used in the LCA. By grounding the research in these expert-provided strategies, this ensures that the quantitative comparisons are heading in the direction of the technically feasible trajectories for the Nordic timber sector.

5.2 Comparative LCA (HT Circular – HT Reference)

This section presents the results of the comparative LCA between HasleTre Circular and HasleTre Reference.

As a reminder, the primary distinction between the two scenarios lies in the connection systems and the approach to modularity. HT Circular utilises reversible wooden connections and modular elements whereas HT Reference employs standard steel connections and non-modular construction methods. The impact of modularity is quantified through its contribution to increased material recovery. Reversible connections contribute to the high reusability rate while simultaneously replacing steel by wood. Table 5.2 summarises these key differences between the two scenarios.

Table 5.2: Summary of the key differences between the two scenarios

HT Circular	**HT Reference**
Reversible wooden connections	Steel connections
Modular elements	Non modular construction
100 % reusability rate of CLT and Glulam	70 % reusability rate of CLT and Glulam

5.2.1. Contribution of the Stages

This section analyses the GWP contribution of each life cycle stage. As illustrated in Figure 5.2, the product stage (A1-A3) is the primary contributor to the total GWP of both building models.

The replacement and refurbishment stages (B4-B5) also exert a significant environmental impact. This is due to the necessity of replacing components with shorter lifespans than the building, such as flooring and windows.

The overall difference in GWP across the entire life cycle remains minor between the two scenarios. This marginal difference is attributed to the fact that the fundamental material composition remains the same, with the only difference being the material of the connections, which are either in wood or in steel.



	A1-A3	A4	A5	B4-B5	C2	C3	C4	D
HT circular	694.9	32.0	85.9	235.1	12.3	36.6	0.1	-681.9
HT reference	695.3	32.0	86.1	235.1	13.0	38.8	0.1	-830.9

Figure 5.2: Graph comparing the GWP by life cycle stage of HT Circular and HT Reference

The impact of these circular design choices can be seen in Module D (benefits and loads beyond the system boundary). Initial results may appear counterintuitive, as Module D shows higher benefits for HT Reference despite its lower reusability rate. This discrepancy is explained by the methodological approach of EN 15804 [55] regarding reuse and energy recovery. Under this standard, net environmental benefits are based on the avoided impacts [56]. When CLT is reused, the credit is based on the avoided production of new timber, which has a relatively low environmental burden. Conversely, when timber is not reused but incinerated for energy recovery, the system receives credits for avoiding the combustion of fossil fuels, which carries a higher environmental impact.

However, it is recognised that the current standard evaluates the life cycle from a linear perspective, which may not accurately reflect the realities of reuse. The current standard credits either the material reuse or its incineration, whereas in reality, these benefits will occur sequentially in the case of reuse. Since a reused component will inevitably reach a final EoL, the benefits should represent the avoidance of virgin material production thanks to reuse, followed by the benefits of energy recovery at the end of its life. When these benefits are aggregated, the advantages of reuse become more pronounced, as illustrated on Figure 5.3. The inclusion of EoL incineration following the reuse phase adds approximately 186 tonnes of CO_2e to module D, which represents an increase of nearly 30 % in the module D benefits compared to the standard calculation method. By failing to

account for this sequence, the standard undervalues the reuse benefits while providing higher credits for immediate incineration. This occurs because the standard is designed for a single-life-cycle analysis and lacks the temporal scope to account for multi-cycle material loops.

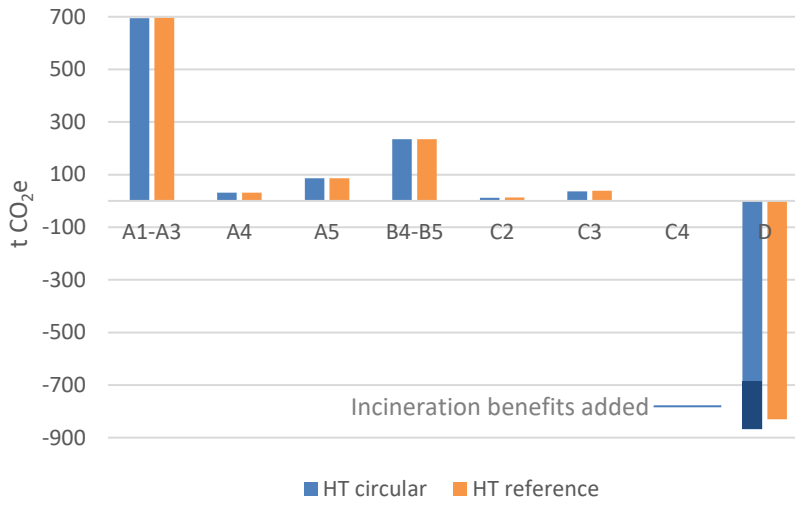


Figure 5.3: Graph comparing the GWP by life cycle stage of HT Circular and HT Reference, with incineration benefits added

The results are further influenced by the fact that biogenic carbon is not explicitly accounted for in the standard GWP results. However, as illustrated in Figure 5.4, the timber products within the project provide a total biogenic carbon storage capacity of over 1000 tonnes of CO_2e. While this carbon is typically released at the EoL through incineration in conventional scenarios, reuse extends the sequestration. In the HT Circular model, approximately 950 tonnes of CO_2e can be stored for a longer duration due to material reuse. In contrast, the HT Reference building, with a lower reusability rate, only retains 690 tonnes of CO_2e. This demonstrate that the circular version allows for almost 30 % more CO_2e to be sequestered over time compared to the reference scenario.

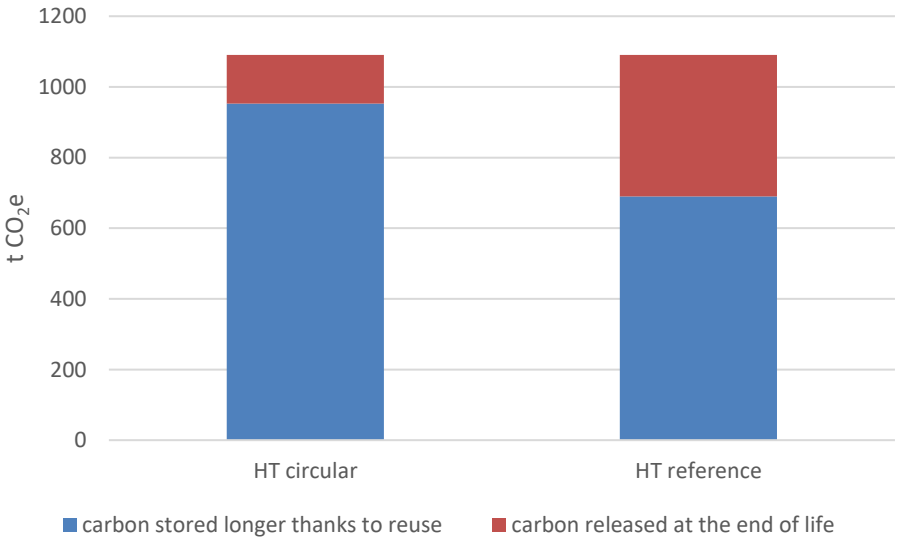


Figure 5.4: Graph comparing the biogenic carbon stored and released between HT Circular and HT reference

5.2.2. Contribution by Material for the Production Stage

In this section, the environmental impacts are analysed by material category for the production stage, as this stage is the most significant one. The primary components, which are consistent in the two building models, are examined first, followed by an evaluation of the connections systems. Although the connections represent a marginal fraction of the total mass and GWP, their role as technical enablers for circularity warrants specific attention.

The relationship between mass and GWP varies significantly across material categories, as illustrated in Figure 5.5.

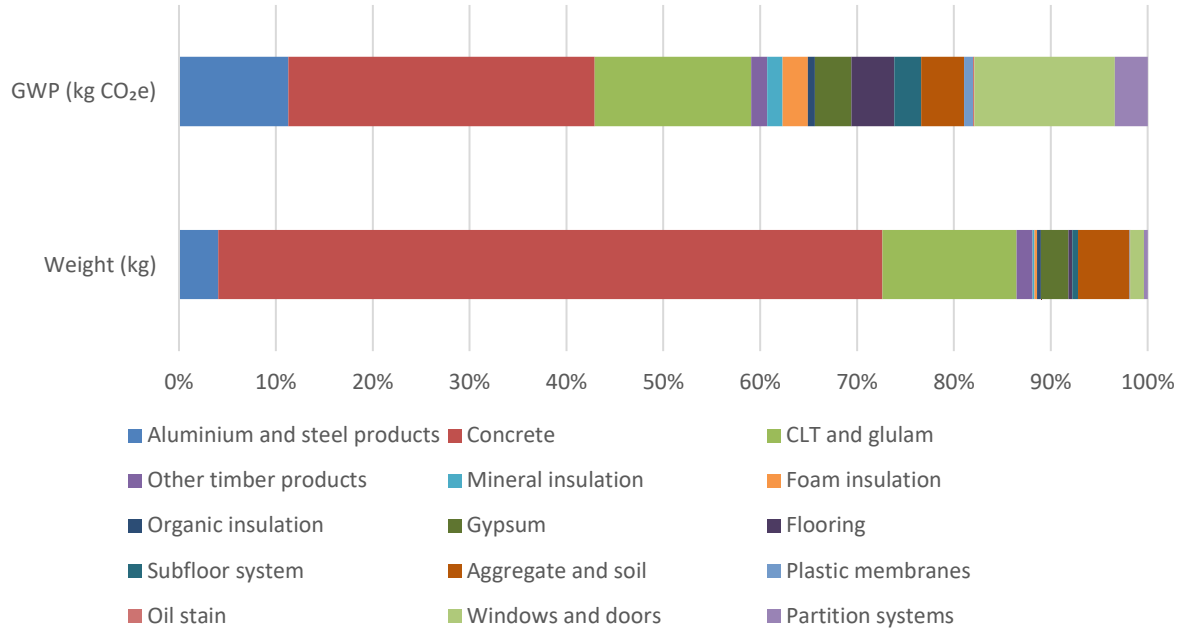


Figure 5.5: Graph of the contribution of the main materials to the total mass and GWP of the production stage (A1-A3)

Concrete represents the largest share of the building mass (68 %) yet contributes only 32% to the total GWP. This disproportionately low impact is a direct result of the use of low-carbon concrete in the project. To quantify this benefit, a simulation was conducted in One Click LCA comparing the project low-carbon concrete with a functionally equivalent standard concrete. The results indicate that the selection of a low-carbon concrete achieved a reduction of approximately 15 tonnes of CO_2e in the production phase, representing a 6 % decrease in the concrete production emissions compared to conventional mixtures.

Steel and aluminium products account only for 4 % of the total mass but contribute 11 % to the GWP. This high impact-to-mass ratio emphasises the necessity of minimising the use of these carbon intensive materials wherever possible.

Windows and doors, despite comprising only 1 % of the total mass, are responsible for 15 % of the total GWP. This is primarily due to their shorter service life compared to the building, necessitating replacement during the building life. Furthermore, high performance windows were used, which are complex and use more materials, resulting in a higher embodied carbon.

Although flooring represents a minimal fraction of the building total mass, it accounts for a disproportionately high share of the total GWP. This is due to the short service life of materials

like carpet, which requires replacement three times over the 50-year lifespan of the building. This finding emphasises that secondary materials, despite their limited volume and weight, can have a substantial influence on the environmental impact due to frequent replacement.

CLT and Glulam account for 14 % of the total mass and 16 % of the total GWP. Given this significant contribution, extending their lifetime through reuse is a positive step in reducing environmental impacts. Furthermore, maintaining these elements in the construction sector allows a long-term biogenic carbon sequestration.

While the connection systems represent less than 0.1 % of the total mass and GWP, they are fundamental to the building circular performance. Reversible wooden connections are evaluated not only for their intrinsic carbon footprint but also for their capacity to facilitate the recovery of large structural components. Yet, it is also worth comparing their inherent carbon footprint with that of conventional steel connections.

As illustrated in Figure 5.6, The X-fix connections are heavier than the conventional steel screws used in HT Reference model. However, their GWP is lower than for the steel, resulting in a saving of 14 kg CO_2e (3 % reduction). Similarly, the comparison of dowel systems (Figure 5.7) favours the wooden dowels in both mass and GWP. The use of wooden dowels rather than steel dowels achieves a saving of 375 kg CO_2e, representing a 98 % reduction. Beyond direct emissions savings, it is important to remember that these wooden reversible connections act as technical enablers, facilitating a higher reusability rate for the mass timber by avoiding metal contamination within the timber.

Furthermore, the HT Reference scenario was modelled with steel dowels as a direct replacement for wooden ones. However, technical discussions with one of the project architects revealed that this simplified substitution does not fully reflect current industry practice. In a traditional 'linear' project, a high-capacity beam-to-column connection would typically require slotted-in steel plates combined with steel dowels. Adopting such a system would not only significantly increase the total mass of steel used but would also compromise the structural integrity of the beams and columns during disassembly, further hindering their potential for high-value-added reuse.

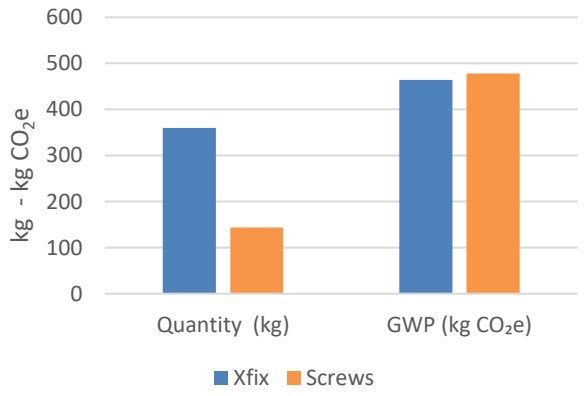


Figure 5.6: Graph comparing X-fix and screws in term of GWP and mass

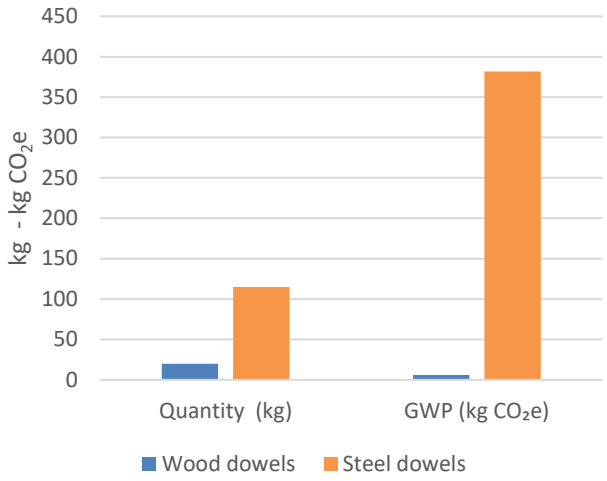


Figure 5.7: Graph comparing wood and steel dowels in term of GWP and mass

5.3 Multi-Cycle Evaluation of CLT Panels

The multi-cycle evaluation focuses on CLT panels, as they represent a primary material in the building construction in terms of both total mass and volume. In the initial whole-building LCA, the CLT ranked third in GWP impact, following metal and concrete, making it the most significant bio-based contributor to the building carbon footprint. Furthermore, these panels are a central focus of the DfD strategy implemented in HasleTre, offering a high potential for high-value structural reuse.

However, for a project such as HasleTre, which is deliberately designed to be dismantled, an analysis focusing solely on a single life cycle will not reveal fully the long-term impact of its structural design. Consequently, a multi-cycle simulation is necessary to determine how the environmental debt should be shared across consecutive uses, as for example the emission form initial production.

The results shown in Figure 5.8 and in Table 5.3 demonstrate how the choice of allocation method alters the perceived environmental impact of the same CLT panels across three life cycles. While the physical emissions and material volumes remain constant, the redistribution and the consideration of the GHG emissions create different profiles for each building iteration.

The Cut-off method (Figure 5.8) is characterised by a heavy concentration of environmental responsibility in the initial life cycle. In this framework, the first building carries the full stage material production impact, making its total GWP significantly higher than the subsequent iterations, and creating an environmental profile that is nearly similar to the linear baseline. However, it does not account for stage C disposal impacts because the material is passed to the next user. Additionally, module D remains zero as the impact of reuse is not yet included. In the intermediate and last life cycles, stage A1-A3 drops to zero, as the material enters the system without material production burden. However, these later cycles now take into account module D, which represents the added environmental burden of the reuse and refurbishment process. The last life cycle finally incorporates the stage C disposal emissions, resulting in a total life impact that is slightly lower than the intermediate building but about three times higher than the initial one, despite using the exact same components.

Under the End-of-Life method shown in Figure 5.8, the production impact is deferred, making the initial cycles appear highly efficient, by crediting the initial building for providing reusable resources. Indeed, in the first and intermediate life cycles, the stage A1-A3 production impact is zero, and stage C disposal is avoided as the panels remain in the loop. These cycles do, however, carry the module D burden associated with impact of making the material reusable for future use. This results in a very low total life impact for the first two iterations. This trend reverses completely in the last life cycle, which bear the weight of the initial product stage and the final disposal stage. Because the material is not reused further regarding this method, module D is zero on this final stage. This method results in a GWP that is three times higher for the final iteration of the building than for the first two. Consequently, the discrepancy between the iterations where production debt is avoided and the one where it is realised is slighter greater in this method than in the Cut-off approach. It can also be noted that this final iteration of the method 0:100 results in an environmental profile nearly identical to the linear baseline iterations, as the deferred environmental debt is finally fully accounted for.

The Distributed allocation method (Figure 5.8) provides a balanced and identical environmental profile for all building iterations by sharing the total impact across the 150-year study period. The stage A1-A3 production debt and the stage C final disposal impacts are divided equally into three parts, ensuring that no single cycle is penalised for the material origin or its eventual end. This method reflects a shared and more stable responsibility model that avoids the extreme fluctuations seen in the other accounting framework but is highly dependent on the number of iterations.

The "without reuse" scenario (Figure 5.8) serves as the baseline, representing a standard linear economy where new CLT is produced for every 50-year building cycle resulting in an identical environmental footprint for each building iteration. Since there is no reuse at the end of each life cycle, the GWP remains high and unchanging across all three cycles. This linear approach requires the full burden of primary production and final disposal to be accounted for every fifty years. It can serve as a benchmark to illustrate the total potential emissions without circular strategies.

The comparative results show that in every case, circularity significantly reduces total emissions over 150 years compared to the linear baseline. Nevertheless, the reporting for the first life cycle remains highly sensitive to the chosen method. As visible in Table 5.3, the GWP for the simulated building varies from 39 to 127 tonnes CO_2e, illustrating how the same structural panels can be characterised as either a high-burden input or a low-impact resource depending on the accounting framework. Furthermore, the inclusion of module D ensures that the technical effort required for deconstruction and refurbishment is consistently captured across all circular scenarios, even as the environmental debt of the primary production is shifted or shared between building iterations.



Figure 5.8: Graph of the GWP of CLT panels reused across 3 building life cycles, using different allocation methods

Table 5.3: Results in tonnes of CO_2 equivalent per life cycle stage for different allocation methods for the CLT panels of HasleTre.

Method	Life Cycle	Module A1-A3	Module A4-A5	Module C	Module D	Total Life Impact
Cut-off w/ Mod D (100:0)	First	104	14	0	0	118
	Intermediate	0	14	0	25	39
	Last	0	14	9	25	48
Avoided burden (0:100)	First	0	14	0	25	39
	Intermediate	0	14	0	25	39
	Last	104	14	9	0	127
Distributed	First	35	14	3	25	77
	Intermediate	35	14	3	25	77
	Last	35	14	3	25	77
Without reuse	First	104	14	9	0	127
	Intermediate	104	14	9	0	127
	Last	104	14	9	0	127

5.4 Sensitivity Analysis

The following section conducts different sensitivity analysis to evaluate the robustness of the Comparative LCA results. Indeed, certain assumptions made in this environmental impact study can influence the final results. By varying one parameter at the time, the implication of these choices can be identified and the validity of the conclusions under different conditions are explored. The analysis specifically examines the impact of building lifespan on the life cycle stages and explores diverse EoL scenarios.

5.4.1. Lifespan

This section presents the findings of the sensitivity analysis comparing building longevity against shorter term structure material reuse. While a 50-year lifespan is the established industry and academic benchmark, as well as the historical LCA practices, this duration was challenged during the FGD [57]. As previously discussed in Section 4.1, experts highlighted a tension between these standard conventions and the technical potential for timber structures to last 100 years or more, notably if designed with flexibility. To address this gap, this sensitivity analysis evaluates whether maintaining a single structure for 100 years is more environmentally effective than deconstructing and rebuilding after 50 years reusing the same timber structure (CLT panels and Glulam beams).

The data illustrated in Figure 5.9 highlights the environmental implications of the two different temporal strategies. While both models utilise DfD, their impact profiles vary significantly across the life cycle stages.

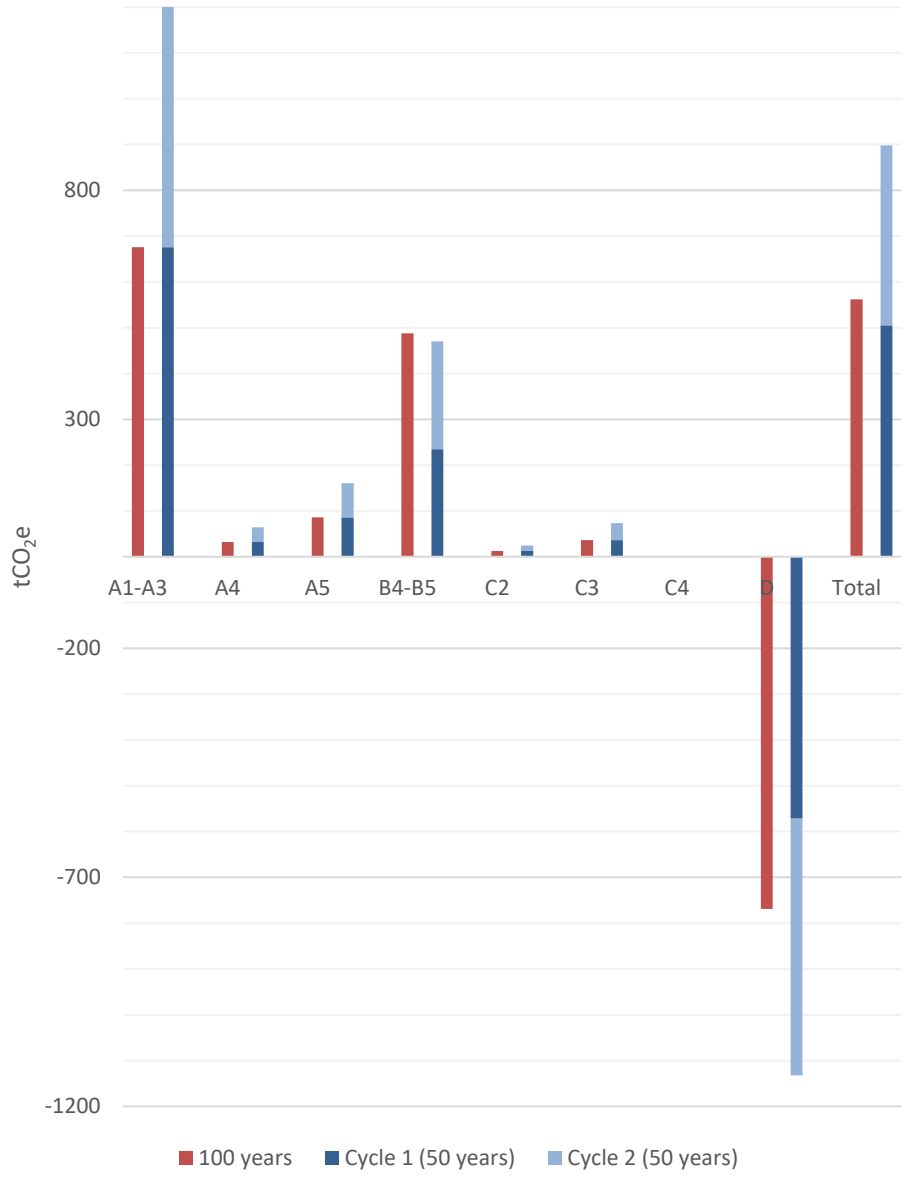


Figure 5.9: Graph comparing the GWP for different life-cycle stages of a circular building with a 100-year lifespan with 2 circular buildings of a 50-year lifespan.

The total impact in the production stage is notably higher for the two-cycle scenario than for the 100-year reference. Although the primary structural element is reused, the second building iterations at year 50 requires a complete set of new secondary components, such as concrete, insulation, membranes, and cladding. Consequently, it is observed that the cumulative scenario has an impact nearly two times higher than the 100-year scenario.

Stages A4 and A5 show a higher cumulative GWP in the multi-cycle scenario due to the repetition of transport and assembly processes for the second building iteration.

In stage B4 and B5, covering replacement and refurbishment, the results are consistent between the two models, as they use similar materials with respective lifespan. The 100-year building requires more replacement cycles over the whole lifespan. In the 50+50 scenario, these replacement

requirements are distributed across two shorter lifespans, resulting in a similar cumulative impact over the total 100-year period.

Regarding module D, an examination of the detailed results reveals that the recovery benefits in module D are influenced by the specific lifespans of the building components. For secondary materials with technical lifespans of 30 to 40 years, the 100-year scenario generates higher cumulative credits because these items undergo more frequent replacement and consequently recovery cycles within the study period. However, this trend is reversed for the primary structure, as the multi-cycle scenario maximises the credit for the CLT panels and Glulam by processing the structural mass through the recovery stage twice, at both the 50-year and 100-year intervals. Consequently, while secondary components favour the longevity model, the total credit for the 50+50 scenario is greater because the structural timber represent an important share of the building materials.

To conclude, despite the high potential for material recovery in the two-cycle system, the total GWP findings indicate that service life extension remains the more effective mitigation strategy. The 100-year reference building results in a total impact of 560 tCO_2e, whereas the circular reuse scenario reaches nearly 900 tCO_2e. The findings support the expert view that maximising the initial service life of a structure is a primary objective for reducing the embodied carbon of the built environment.

However, it is important to note that this model only accounts for the reuse of the primary timber structure, assuming all secondary components are replaced. Recovering a broader range of elements could reduce the production, especially as the building was built for disassembly including dry-assembled layers. Additionally, while the building is assumed to be well constructed, the model does not account for potential material degradation that might occur over the planned lifespan of CLT panels and Glulam.

5.4.2. End-of-Life Scenarios

This section evaluates how varying the EoL scenarios for the non-reused portions of CLT and Glulam impacts the LCA results. While the initial comparative LCA (Section 5.2) showed a paradoxically higher Module D benefit for HT Reference due to energy recovery credits, this analysis investigates whether reuse provides superior benefits when compared to other common EoL like landfilling or informal recycling.

As the variables modified concern only the treatment of the non-reused material fractions (the 30 % loss in HT Reference), the impacts for stages A1 to C4 remain relatively stable at the building scale across scenarios, as illustrated in Figure 5.10. The primary difference occurs in Module D. When compared to landfilling or informal recycling, HT Circular shows higher benefits. This highlights that the counterintuitive results observed in the initial comparison, described in Section 5.2, were a specific consequence of high credits given to wood incineration for replacing fossil-fuel in district heating.

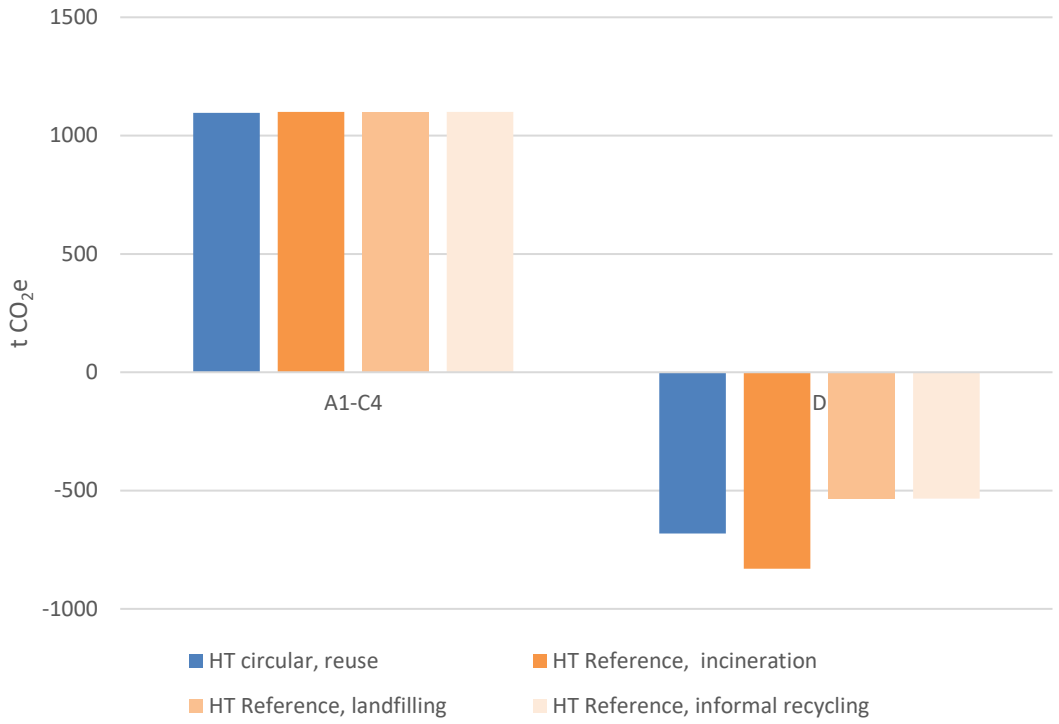


Figure 5.10: Graph comparing the GWP of different End-of-Life scenarios

To isolate and analyse the impacts at different life cycle stages, Table 5.4 focuses specifically on the 30 % of CLT and Glulam components that vary across the different EoL scenarios. The production (A1–A3) and transport (A4) impacts remain identical for all scenarios, as seen in Table 5.4.

Table 5.4 GWP in kg CO_2e of the non-reused fraction of CLT and Glulam, with different EoL scenarios, by life cycle stage

	HT circular, reuse	HT Reference, incineration	HT Reference, landfilling	HT Reference, informal recycling
A1-A3	33890.8			
A4	870.0			
A5	3476.1	3767.7	3583.2	870.0
C2	0	669.2	267.7	267.7
C3	0	2247.1	0	2247.1
C4	0	0	803.7	0
D	-37279.9	-186334.6	0	0

Figure 5.11 highlights the changes occurring in the stages of construction (A5), transport during the EoL (C2), waste processing (C3) and disposal (C4).

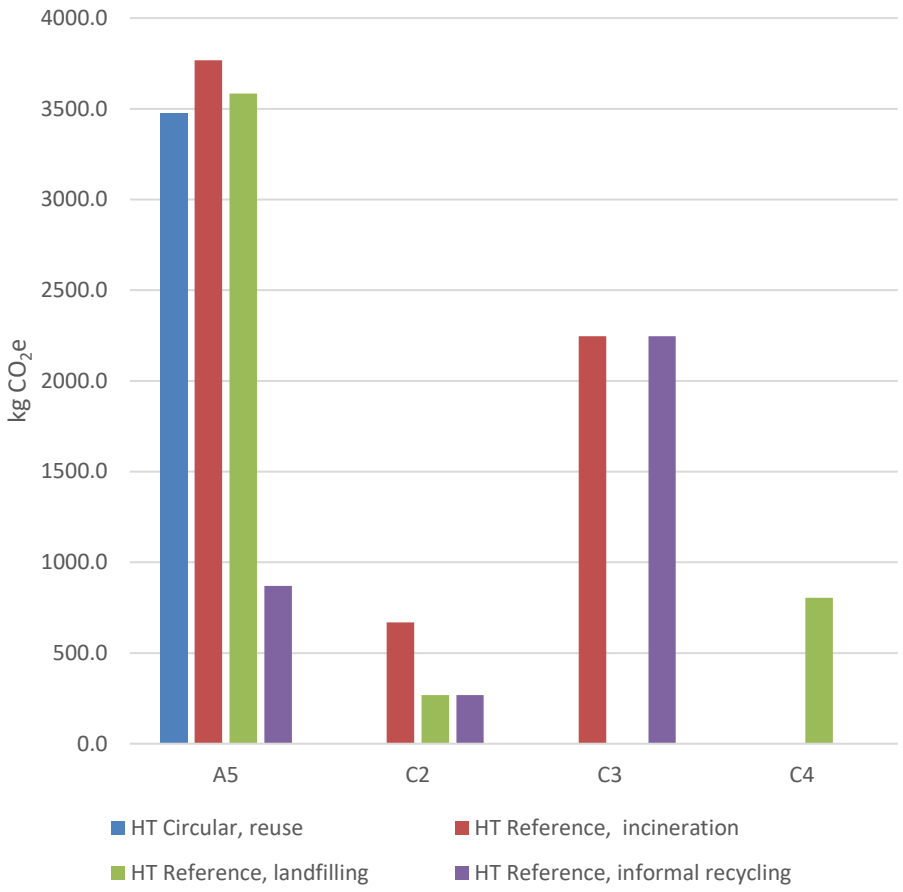


Figure 5.11: Graph comparing the GWP of the non-reused fraction of CLT and Glulam, with different End-of-Life scenarios, by life cycle stage (excluding the constant ones and Module D)

For reuse (HT Circular), impacts are primarily in the stage of construction (A5) due to onsite material wastage. The stages concerning the EoL, i.e. transport (C2), waste processing (C3) and disposal (C4) are zero because the materials remain within a closed loop, with loads of dismantling, transport and reassembly accounted in Module D.

Incineration (HT Reference) shows impacts in the transport stage (C2) and in the waste treatment stage (C3) which, in this case, is incineration. Although the wastage in construction is the same as for reuse, the environmental impacts of construction (A5) are higher because that waste is also processed through incineration.

Landfilling (HT Reference) has a lower transport impact (C2), likely due to the higher geographic availability of landfill sites compared to specialised treatment plant such as for incineration. However, it increases significantly the disposal impacts (C4).

Finally, informal recycling (HT Reference) assumes that waste is recovered outside formal systems (e.g., taken by private users). It shows the lowest construction impacts (A5), as the burden of waste management is shifted away from the project.

While a quick look at Figure 5.11 might suggest landfilling is a good option, this is a misleading conclusion that ignores Module D benefits. As shown in Table 5.4, landfilling and informal recycling provide zero environmental credits. Landfilling offers no recovery potential, neither for

energy or material, and informal recycling, while beneficial by avoiding virgin material production, cannot be formally credited in an LCA because it occurs outside of formal systems.

This sensitivity analysis demonstrates that the superior Module D results for the HT Reference building compared to HT Circular, are a direct consequence of the methodology for giving credits to wood incineration. When reuse is compared against alternative EoL pathways such as landfilling or informal recycling, it emerges as the most environmentally beneficial option. While Module D remains a valuable tool for quantifying the benefits of circular design, a significant methodological challenge persists: the current framework overvalues incineration credits, which can obscure the true climate benefits of material reuse.

6 Discussion

This chapter provides a critical evaluation of the research by synthesising the findings within a broader context and analysing the limitations and strength of the chosen approach. It is structured into two main subsections: the first section discusses the results, while the second evaluates the methodology.

6.1 Discussion of the Results

This section interprets and contextualises the environmental implications of the HasleTre case study by connecting the qualitative expert perspectives with the quantitative results of the LCAs and sensitivity analyses.

6.1.1. Materials

Regarding the materials, the results of the comparative LCA reveals that the total GWP difference between the circular (HT circular) and linear (HT Reference) designs is relatively minor when viewed across the entire 50-year life cycle. This marginal variation is best understood by acknowledging that the fundamental material composition of both buildings is nearly identical. The primary distinction lies in the connection system and the theoretical recovery rates.

In both scenarios, the product stage (A1-A3) accounts for two-thirds of the building total GWP, i.e. the majority. This dominance of the material production phase aligns closely with other studies. For instance, a Swedish case study evaluating a multi-storey CLT building similarly demonstrated that the product stage accounted for 68 % of the total GWP when the use stage was excluded [28].

As a reminder, concrete represents the largest share of the building mass at 68 % yet contributes 32 % of the production-related GWP. In contrast, aluminium and steel products contribute 11 % of the production GWP despite making up only 4 % of the mass. Collectively, the results shows that these traditional materials dominate roughly the impact from stages A1 to A3, while the CLT and Glulam structure accounts for 16 %. Consequently, the circular strategies addressing wood structure and connections focus on a smaller, though significant, fraction of the total carbon footprint.

Concerning concrete, its share in the building production GWP impact may appear surprisingly low. In this project, the specified low-carbon concrete has a lower impact per kilogram ratio than the timber elements. However, this comparison requires critical context to avoid misinterpretation. First, this analysis excludes biogenic carbon sequestration, a factor that would improve the net benefits of the timber components. Second, the functional role of the material differs drastically: while the timber volume is sufficient to frame the entire four-storey structure, the concrete is strictly limited to the basement and foundations. As highlighted in several comparative studies of wood and concrete structures from the literature, achieving equivalent structural performance requires three to five times more mass when using concrete compared to mass timber elements such as Glulam and CLT [34], [58]. Consequently, even if the impact per kilogram of concrete appears lower, its total environmental burden for an equivalent building would be higher due to the greater volume and mass required. Furthermore, as noted during a discussion with an architect from Oslotre, the design phase prioritised the DfD and circularity of the timber structure, resulting in less emphasis placed on finding alternatives to concrete for the sub-structure.

Beyond structural elements, the production GWP-to-mass ratio is also high for specific finishes and envelope components such as windows and floor coverings. Triple-glazed windows, for example, are a frequent choice for high-performance buildings in this climate, to minimise

operational energy demand. Depending on standards and regulations, it may be worthwhile to compare their impact with double glazed or other types of windows, to assess whether the additional embodied carbon is worth it considering the operational energy savings that can be achieved. Similarly, the use of wool carpets with a 20 % polyamide content provides acoustic insulation and durability for an office environment. During discussions, an architect from Oslotre noted that the proportionally high impact of these carpets was not immediately apparent during the design phase, highlighting how functional requirements like comfort and acoustics can sometimes compete with carbon reduction goals.

This underscores that material selection is rarely dictated by a single metric. A holistic approach requires balancing environmental impact with technical performance, cost, and occupant well-being. Indeed, creating an aesthetically pleasing and comfortable environment is also a factor in extending a building service life. A space that people enjoy occupying is more likely to be maintained and preserved over time. Ultimately, the choice of materials involves a complex compromise where GWP must be weighed against the functional necessity of the components and the practical constraints.

These observations, while specific to the HasleTre case study, suggest a broader trend in contemporary sustainable architecture, which is the difficulty of addressing all environmental factors simultaneously. While the building is a pioneer in timber circularity, the reliance on concrete foundations illustrates that even high-ambition projects must often prioritise specific areas of innovation. This highlights that achieving a holistic low-carbon approach across all building systems remains a complex challenge for the sector.

6.1.2. Connections

While the LCA results shows that reversible wooden connections (X-fix and dowels) account for less than 0.1 % of the total mass and GWP, their importance seems to be strategic rather than volumetric. In the FGD, experts prioritised modular elements and reversible systems as the top technical solutions for circularity in timber buildings. From a quantitative perspective, replacing steel fasteners with wood achieves a direct saving of 3 % CO_2e for the X-fix and 98 % for the dowels. However, it is important to remember that the primary value of these joints lies in their ability to act as enablers for the larger structure. As described in the literature review, traditional circular strategies are often affected by technical challenges like adhesives and nails that complicate deconstruction. By utilising all-wood joints, HasleTre mainly avoids the risk of embedded steel, which allows the model to theoretically increase structural timber recovery from 70 % to 100 %.

Nevertheless, this relies on the significant assumption that the structural integrity of the CLT panels is maintained perfectly over time. This assumption is needed according to the knowledge gap identified in the literature regarding the long-term durability of mass timber. Added to this is the lack of data also on the long-term behaviour of reversible connections like X-fix and wood dowels, as mentioned in the FGD. This therefore leaves many unknowns regarding the suitability of reversible connections for reusing timber structural elements. However, they can still be viewed as "no regret" strategy. In the worst case, even if structural reuse is not realised, their implementation does not incur a higher environmental burden than traditional metal fasteners. Furthermore, X-fix connectors provide other advantages. For example, they are installed manually and aesthetically pleasing, which means they do not need to be concealed under cladding, as it is the case with metal fasteners. It should be noted, however, that price is also a factor to be taken into account, and may put wood connections at a disadvantage, as they are not yet widely available.

6.1.3. Module D Paradox

In this research, the current main difference between the HT circular and HT Reference lies in module D and its paradox. The linear HT Reference building showed higher environmental benefits because the current EN 15804 [55] standards provide high credits for wood incineration, which displaces fossil fuels in district heating. As identified in the sensitivity analysis, this result was counterbalanced by demonstrating that other EoL processes make reuse more advantageous. This paradox is clarified by evaluating the cumulative impact, aggregating the benefits of material reuse with the energy recovery from final incineration. By accounting for these sequential stages, it becomes clear that the circular approach provides superior environmental benefits that is otherwise undervalued by the assessment following the standard. This finding supports the necessity for a methodological shift that captures the full temporal scope of multi-cycle material loops.

This module D paradox also highlights the issue of biogenic carbon accounting mentioned in the literature. While GWP results might for now favour incineration, the results show that the circular model allows for 30 % more CO_2e to be sequestered by extending the material lifespan. Although this study does not account for this in its life-cycle analysis, and despite this rate being based solely on wood quantity, its inclusion could provide an opportunity to mitigate the results. Prioritising reuse prolongs carbon storage, offering a much favourable alternative to incineration and is associated CO_2 emissions.

These inconsistencies in the results and the complexity of calculating biogenic carbon, as explained in the literature review, clearly demonstrate the importance of generalising and improving the way reuse is accounted for in calculation tools. This ties in with the need for a proper framework for managing reused materials, as outlined above. The reuse of materials would benefit from a more clearly defined theoretical and practical framework based on circular rather than a linear representation.

The necessity of a regulatory framework is not only a matter of construction standards but also of accounting transparency. Indeed, if the sector aims to make reuse attractive to customers, both in the design and use of reused raw materials, how it is accounted for is a key issue. For now, the perceived environmental success of a building remains highly sensitive to the chosen LCA methodology.

However, as the multi-cycle simulation of the HasleTre CLT panels demonstrates, this "sharing" of impact is far from standardised. The drastic fluctuation in GWP results, ranging from 39 to 128 tonnes of CO_2e for the same physical structure, highlights that the perceived sustainability of a circular building is currently dependent of methodological choice. This closely aligns with the findings of De Wolf et al. [39], whose evaluation of different allocation methods similarly concluded that the final interpretation of environmental performance is fundamentally dictated by the chosen allocation framework.

This methodological disparity directly influences stakeholders strategic decisions based on their circular goals. For instance, a practitioner focusing on DfD would favour an EoL allocation framework that rewards the building for enabling a higher theoretical reusability rate, providing significant credits for future life cycles. Conversely, an actor incorporating reused materials into a new project requires a framework that recognises the immediate displacement of primary production emissions (module A1-A3) like the cut-off method, safeguarding components that would otherwise be downgraded into standards EoL process. Consequently, the timing of environmental credits under different allocation methods can either incentivise or penalise the initial investment in high-quality, reusable timber structures.

6.1.4. Extending the Lifespan

Although the literature review and the results of this study highlighted the importance of EoL considerations and reuse of CLT, a comment made during the FGD was worth exploring. Indeed, the conventional 50-year lifespan was questioned with the argument that buildings should be designed to last at least 100 years. It was added that Design for Flexibility (DfF) is in that case essential, as it allows the building to adapt to changing needs across the time, and directly support longer lifetime. According to one of the architect for the project, this type of design does not require more materials, but rather thoughtful planning from the very start of the project. Furthermore, DfD, which is promoted as a way of facilitating EoL management and reuse, also plays a role in extending the building lifespan. For example, dry-assembled layers also facilitate the replacement of materials that have a shorter lifespan than the building itself, allowing a focus on repairing rather than rebuilding.

The sensitivity analysis supports the experts insights by showing that extending service life is superior to material circularity across multiple buildings. The 100-year longevity scenario resulted in a total impact of approximately 560 tonnes of CO_2e, compared to nearly 900 tonnes of CO_2e for the two-50-year longevity scenario where the structure is reused.

A literature review on the EoL pathways of mass timber products by Lin et al. [59] corroborates these findings. The authors establish that the most effective EoL strategy is to delay carbon release by extending the lifespan of the entire building structure to avoid demolition, while the direct reuse of mass timber elements represents the second most favourable scenario.

This reinforces the expert view that, while circularity is vital contingency, the main environmental goal must remain the extension of the service life to minimise the high carbon costs of new construction phases. Nevertheless, as was highlighted during the FGD, Design for Circularity requires support and commitment from the client to build to standards that go beyond the minimum cost. This is also in line with the conclusions of the preceding paragraphs, which call for the development of a framework to promote and regulate the practices described here, so that they become the norm rather than an advantage.

6.1.5. Frameworks, Certification, and Ecosystem Readiness

Sustainability certification schemes like BREEAM, LEED, Miljöbyggnad or FutureBuilt Zero often serve as the primary benchmarks for sustainable building in the current market. While a detailed study of these certifications was beyond the scope of this research, our observations suggest that these frameworks play a significant role in defining which environmental aspects are prioritised. For instance, an LCA for the HasleTre project was conducted by an external consultant to meet FutureBuilt and BREEAM criteria. A preliminary analysis of this LCA revealed a specific, non-holistic approach that did not account for all building elements or life cycle stages. While such omissions are likely justified by the certification specific scope and requirements, they highlight the risk that projects may be designed primarily to satisfy specific benchmarks rather than achieve global sustainability. Also, if the scheme used does not reward or require the accounting of long-term material recovery and biogenic carbon storage, the efforts invested in modularity and disassembly might not be fully reflected in the final environmental assessment. Ultimately, while these certifications are key drivers that make sustainable construction valuable on the market and provide helpful accounting guidelines, it remains essential to maintain a holistic long-term vision for the building impact.

Optimising a building on paper within a certification framework is only the first step, as real-world recovery also depends on the external ecosystem capable of handling, dismantling, transporting,

refurbishing, and storing, as well as matching these materials with new construction projects. Currently, infrastructure like Sirkulær Resurssentral in Norway, or digital platforms like CCBuild in Sweden provide the necessary logistics and storage space to bridge the gap between deconstruction and new construction as described in the Literature Review and Theory [60], [61]. Despite this emerging infrastructure, a hurdle remains regarding structural guarantees. It is difficult to assume that reusable CLT or Glulam has the same load-bearing capacity as a brand-new material. Or, if that is the case, contractors and engineers may need assurance that this is indeed the case. These issues, which relate to the non-destructive testing of structural components, lie outside the scope of this study. However, they illustrate the diversity of stakeholders and levels involved in the reuse sector as well as the scope for improvement in this area.

To conclude, the evaluation of the HasleTre case study suggests that, while circularity might be technically achievable through modularity and reversible connections, its actual environmental performance appears dependent on a complex chain of factors. Beyond the initial design, effective reuse is contingent on the quality of the construction and the aging process of the materials, as well as the development of external logistics for refurbishment, testing, and storage. Furthermore, a shift in both the regulatory framework and the mindset of clients and designers is necessary for these systems to function. However, the findings suggest that the priority to improve overall GWP impact of wooden building remains the extension of the service life of buildings. While the LCA findings provide a quantitative baseline, they also suggest that the perceived success of a sustainable project could be influenced by carbon accounting choices and market valuation of future material potential. This reliance on specific accounting parameters necessitates a critical evaluation of the methodological framework and the assumptions that supported the quantitative results of this study, which is presented below.

6.2 Discussion of the Method

After discussing the results of this study, several methodological factors must be discussed to contextualise these findings and acknowledge the gap between the modelled LCA and the actual complexity of the HasleTre building.

A primary technical consideration in this study is the exclusion of biogenic carbon from the primary GWP results. As highlighted by Dodoo et al. [20], static GWP models fail to represent the complex behaviour of carbon sequestration and release in timber structures. The results of their study indicate that when these biogenic carbon flows are not adequately considered, the potential for timber buildings to mitigate climate change is typically reported as lower than it is in reality. By keeping wood-based materials in the loop for multiple lifecycles, HasleTre functions as a long-term carbon pool. If a dynamic LCA approach had been applied, the benefits of the HasleTre Circular scenario would likely be higher, as the storage of biogenic carbon would be credited.

In addition to these carbon dynamics, the reliability of any LCA is fundamentally dependent on the data used. In this study, while material quantities were provided by the architects, full specifications for all products were not available. Assumptions therefore had to be made regarding material types and the selection of EPDs when exact data was missing. While this introduces uncertainties, these assumptions are acceptable for a comparative LCA because they were applied identically to both scenarios for all materials except connections. Regarding the connections, the modelling of X-fix connectors and beech dowels required a geometric conversion into material volumes, due to a lack of specific environmental data about them, which may not capture the complete industrial manufacturing process, and the total carbon footprint associated with these specific components.

Beyond the data constraints of the model, it must also be acknowledged that this LCA captures only a fraction of the circular strategies implemented in the actual HasleTre project. As described in Section 3 (Case Study Description), this project incorporated numerous measures to ensure a sustainable and long-lasting building, such as the creation of flexible floor plans or easy access to techniques. Furthermore, salvaged materials such as sanitary or furniture were integrated in the design. However, these efforts were not fully captured by the LCA, which focused specifically on the connection systems and the increased reusability rate. It remains difficult to quantify the comprehensive design reflections behind the project, even though they significantly contribute to its overall sustainability. Consequently, while the results demonstrate environmental benefits, they likely underestimate the true positive impact of the building.

This gap between modelled data and real building is further influenced by the specific framework chosen for the assessment. The use of One Click LCA as the modelling tool was motivated by its extensive database of EPDs, yet the study followed the EN 15804 +A1 [62]s tandard. While this provides a regulated framework for evaluating the environmental performance of buildings, it is not the most recent version (EN 15804 +A2 [63]) . The GWP in the standard used here does not incorporate biogenic carbon, which, while in line with the chosen methodology for this study, contributes to the conservative nature of the results. In addition, while the EN 15804 +A1 [62] standard only mandates the reporting of product stage impacts (A1-A3), this assessment expanded the system boundary to include construction, use and EoL stages. The inclusion of these modules allows a more comprehensive cradle-to-gate perspective. However, it necessitated the use of assumptions from the One Click LCA database to supplement the modules not covered in specific EPD's, which introduces a degree of uncertainty compared to using exclusively manufacturer-verified data.

Furthermore, the reliance on GWP as the sole environmental indicator to quantify the benefits of the building presents its own limitations. While GWP is a widely used metric for assessing climate change impacts, isolating this single indicator may lead to a shift of burden, where improvements in GHG emissions inadvertently result in higher environmental impacts in other impact categories. It is thus important to keep in mind that these results only show a portion of the total environmental impacts.

Completing these quantitative models, the study integrated qualitative insights, which must also be viewed critically. The FGD served as an important qualitative tool to bridge the gap between theory and practice, providing valuable insights into the current realities of the construction sector. However, it must be acknowledged that the qualitative data derived from this session is inherently subjective. The participants were recruited based on their expertise and existing engagement with CE principles, which resulted in a group already convinced of the benefits of circular strategies. Consequently, the consensus and rankings reached during the FGD likely represent an optimistic outlook rather than a consensus of the broader and more conservative construction sector. Furthermore, a standard FGD process should ideally consist of multiple meetings with different groups, whereas this study relied on a single session, making the results more subjective.

Finally, all discussed factors are situated within a specific geographic context. The findings of this research are contextualised within the Nordic region, specifically Norway and Sweden. This scope is significant because the electricity grids in these countries are largely decarbonised, which increases the relative importance of embodied carbon over operational emissions. Additionally, timber construction is culturally and technically integrated in these countries. These factors mean that the findings of this research may not be directly transferable to regions with different material availability or energy mixes.

7 Conclusions

Bringing together the entire research process, this section summarises the key findings and present the conclusions of the study, evaluates its applicability to the sector, and identify key areas for future research.

The primary objective of this thesis was to evaluate the environmental impacts of reusing CLT and Glulam elements with wooden reversible connections in the HasleTre project, located in Oslo, Norway. Through a mixed-methods approach combining qualitative expert insights with quantitative LCAs, it was demonstrated that while circular strategies involve minimal changes to the total building mass, they serve as a critical enabler for long-term climate mitigation.

The investigation revealed that the substitution of metal connections with wooden alternatives does not significantly reduce the total production-related GWP, as connections represent less than 0.1 % of the total building mass and production-related GWP. However, these connections are fundamental in increasing the theoretical reusability rate of structural components from approximately 70 % to a range approaching 100 %. It was observed that module D results can be counterintuitive under current standards, as wood incineration for energy recovery receives higher credits than material reuse. Indeed, the HT Reference building received higher credits in Module D, with -830 tonnes of CO_2e, than the HT Circular building, with -680 tonnes of CO_2e. Nevertheless, sensitivity analyses of alternative EoL scenarios confirmed that reuse remains the most environmentally beneficial trajectory. These findings underscore that current LCA frameworks are not yet adapted to represent the complexities of multi-use cycles.

Furthermore, the multicycle evaluation demonstrated that while reuse offers clear environmental advantages, the distribution of these benefits is highly dependent on the chosen allocation method. The results varied significantly, with total GWP values ranging from 39 to 127 tonnes of CO_2e for the same CLT panels, depending on the accounting method. This variation is primarily due to how different methods distribute the initial manufacturing carbon burdens and subsequent EoL credits between the first building and future product life cycles. This discrepancy highlights a critical need for standardised frameworks to ensure consistency in circular accounting.

Ultimately, this study concluded that building longevity remains the most effective environmental strategy for reducing the total environmental impact, as suggested by the wood construction experts taking part in the FGD. The 100-year longevity scenario resulted in a total impact of approximately 560 tonnes of CO_2e, compared to 900 tonnes of CO_2e for the scenario where the building is deconstructed and rebuilt with the same structural components. However, circular design provides the essential capacity for a building to be deconstructed, and its materials reclaimed when the initial service life is terminated.

These findings emphasise the critical need for evolving LCA standards to better reward material reuse, ensuring that the true climate benefits of timber circularity are accurately reflected in the building sector.

7.1 Applicability of the Study

This research may serve as a resource for architects, structural engineers, and developers exploring the implementation of DfD in mass timber buildings. Specifically, the data on mass timber modular elements and reversible wooden joints provides designers with an empirical example of structural recovery pathways. Furthermore, this work can help practitioners anticipate how non-structural choices, such as the building envelope or interior finishes, affect the overall environmental footprint. It supports a more holistic approach during early-stage design planning.

At a systemic level, the analysis of the module D paradox and the multi-cycle simulations might be of interest to policymakers, certification bodies, and LCA practitioners. By illustrating how GWP results vary depending on the chosen allocation method, this study provides a contextual example that could inform ongoing professional discussions regarding the limitations of single-life-cycle frameworks. Consequently, it may assist in the future development of carbon accounting methodologies and sustainability benchmarks for circular construction.

7.2 Future Work

While this thesis validates the environmental potential of circular timber design, several areas remain for further investigation.

First, biogenic carbon was excluded from the scope of this study, which likely resulted in a more conservative reporting of the building environmental performance. To more accurately reflect the climate benefits of mass timber circularity, future studies should implement dynamic LCA methodologies. Such an approach would reward the temporal benefits of delayed carbon emissions achieved through the reuse of timber elements.

Secondly, while wooden connections were assumed to improve the ease of deconstruction and help increase the reusability rate, research is needed to evaluate the structural durability and non-destructive removal of X-fix and wooden dowel connections after extended periods of building service life, specifically considering the effects of wood moisture fluctuations and creep.

Thirdly, the economic feasibility of reversible wooden connections compared to conventional metal fasteners is worth investigating. At present, these timber-to-timber joints may incur higher upfront cost because they represent a relatively new and not yet widely adopted technology requiring high precision during factory pre-milling. Future research should incorporate a life cycle costing analysis to evaluate whether these initial costs are eventually offset by faster on-site assembly, reduced deconstruction labour, and the preservation of the residual value of the components.

Finally, there is a critical need to develop a standardised legal and technical framework for the structural re-certification of salvaged CLT and Glulam to provide the structural guarantees required for future construction.

References

Generative AI tools (Gemini and Notebook LM) were used during the preparation of this thesis. Specifically, AI was used to assist in the preliminary identification of potential academic literature for the review; however, all sources were then reviewed and verified by the authors. Additionally, AI tools were employed to refine the linguistic quality, clarity, and flow of the text in English. All final results, critical analysis, and conclusions were generated solely by the authors, who hold full accountability for the content of this work.

[1] United Nations Framework Convention on Climate Change (UNFCCC) Secretariat, 'The Paris Agreement | UNFCCC'. Accessed: Mar. 03, 2026. [Online]. Available: https://unfccc.int/process-and-meetings/the-paris-agreement

[2] A. Sáez-de-Guinoa, D. Zambrana-Vasquez, V. Fernández, and C. Bartolomé, 'Circular Economy in the European Construction Sector: A Review of Strategies for Implementation in Building Renovation', *Energies*, vol. 15, no. 13, p. 4747, Jan. 2022, doi: 10.3390/en15134747.

[3] Eurostat, 'Generation of waste by economic activity'. Eurostat, 2022. doi: 10.2908/TEN00106.

[4] Statistikmyndigheten SCB, 'Waste generation', Statistikmyndigheten SCB. Accessed: Mar. 03, 2026. [Online]. Available: https://www.scb.se/en/finding-statistics/statistics-by-subject-area/environment/waste/waste-generated-and-treated/pong/tables-and-graphs/waste-generation/

[5] Statistisk sentralbyrå (SSB), 'Waste accounts', SSB. Accessed: Mar. 03, 2026. [Online]. Available: https://www.ssb.no/en/natur-og-miljo/avfall/statistikk/avfallsregnskapet

[6] G. L. F. Benachio, M. do C. D. Freitas, and S. F. Tavares, 'Circular economy in the construction industry: A systematic literature review', *Journal of Cleaner Production*, vol. 260, p. 121046, Jul. 2020, doi: 10.1016/j.jclepro.2020.121046.

[7] European Parliament, 'Circular economy: definition, importance and benefits', Topics | European Parliament. Accessed: Mar. 03, 2026. [Online]. Available: https://www.europarl.europa.eu/topics/en/article/20151201STO05603/circular-economy-definition-importance-and-benefits

[8] Circle Economy Foundation and RISE Research Institutes of Sweden, 'CGR Sweden'. Accessed: Mar. 03, 2026. [Online]. Available: https://www.circularity-gap.world/sweden

[9] Circle Economy Foundation, Skift Business Climate Leaders, and Greenpeace Norway, 'Norway - Circularity Gap Reporting Initiative'. Accessed: Mar. 03, 2026. [Online]. Available: https://www.circularity-gap.world/norway

[10] European Commission, *Communication from the commission to the European parliament, the council, the European economic and social committee and the committee of the regions: Closing the loop - An EU action plan for the Circular Economy*. 2015. Accessed: Mar. 03, 2026. [Online]. Available: https://eur-lex.europa.eu/legal-content/en/TXT/?uri=CELEX%3A52015DC0614

[11] European Parliament and the Council of the European Union, *Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008 on waste and repealing certain Directives (Text with EEA relevance)*, vol. 312. 2008. Accessed: Mar. 03, 2026. [Online]. Available: http://data.europa.eu/eli/dir/2008/98/oj

[12] European Parliament and the Council of the European Union, *Regulation (EU) 2020/852 of the European Parliament and of the Council of 18 June 2020 on the establishment of a framework to facilitate sustainable investment, and amending Regulation (EU) 2019/2088 (Text with EEA relevance)*, vol. 198. 2020. Accessed: Mar. 03, 2026. [Online]. Available: http://data.europa.eu/eli/reg/2020/852/oj

[13] European Parliament and the Council of the European Union, *Regulation (EU) 2024/1781 of the European Parliament and of the Council of 13 June 2024 establishing a framework for the setting of ecodesign requirements for sustainable products, amending Directive (EU) 2020/1828 and Regulation (EU) 2023/1542 and repealing Directive 2009/125/EC (Text with EEA relevance)*. 2024. Accessed: Mar. 03, 2026. [Online]. Available: http://data.europa.eu/eli/reg/2024/1781/oj
[14] CIRCuIT Project, 'Futureproofing cities: designing for disassembly and adaptability | Circular construction in regenerative cities'. Accessed: Mar. 03, 2026. [Online]. Available: https://report.circuit-project.eu/chapter/futureproofing-cities-designing-for-disassembly-and-adaptability
[15] J. A. Myhren, A. Dodoo, and F. Quarcoo, 'Challenges In the Transition To Circularity In Multistorey Wood Building Construction : Scandinavian Industry Perspective', in *14th World Conference on Timber Engineering (WCTE 2025) : Advancing Timber for the Future Built Environment.*, Brisbane, Australia, 2025, pp. 4601–4607. Accessed: Oct. 21, 2025. [Online]. Available: http://urn.kb.se/resolve?urn=urn:nbn:se:lnu:diva-142113
[16] United Nations Environment Programme and Yale Center for Ecosystems + Architecture, *Building Material and the Climate: Constructing a New Future*. United Nations Environment Programme, 2023. doi: 10.59117/20.500.11822/43293.
[17] T. Malmqvist, M. Erlandsson, N. Francart, and J. Kellner, *Minskad klimatpåverkan från flerbostadshus – LCA av fem byggsystem*. 2018. Accessed: Mar. 03, 2026. [Online]. Available: https://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-298399
[18] A. Amiri, J. Ottelin, J. Sorvari, and S. Junnila, 'Cities as carbon sinks—classification of wooden buildings', *Environmental Research Letters*, vol. 15, no. 9, p. 13, Aug. 2020, doi: 10.1088/1748-9326/aba134.
[19] G. Churkina *et al.*, 'Buildings as a global carbon sink', *Nat Sustain*, vol. 3, no. 4, pp. 269–276, Jan. 2020, doi: 10.1038/s41893-019-0462-4.
[20] A. Dodoo, R. Sathre, and C. Piccardo, 'Climate impacts of timber-frame buildings in a life cycle perspective: Implications of static and dynamic modelling approaches for biogenic carbon accounting', *Environmental Impact Assessment Review*, vol. 117, p. 108213, Mar. 2026, doi: 10.1016/j.eiar.2025.108213.
[21] J. Swartz Andersen *et al.*, 'Economic Comparison of Mass Timber and Concrete Construction in The Nordic Region', in *World Conference on Timber Engineering (WCTE 2023)*, Oslo, Norway: World Conference on Timber Engineering (WCTE 2023), 2023, pp. 4360–4369. doi: 10.52202/069179-0568.
[22] Swedish Wood, *The Glulam Handbook - facts about glulam*, vol. Volume 1, 3 vols. Stockholm, 2024.
[23] Wooclap, *Wooclap*. (2026). Wooclap. Accessed: Mar. 05, 2026. [Online]. Available: https://www.wooclap.com
[24] Ember, 'Lifecycle carbon intensity of electricity', Our World in Data. Accessed: Mar. 23, 2026. [Online]. Available: https://ourworldindata.org/grapher/carbon-intensity-electricity
[25] A. Laurent, S. Olsen, and M. Hauschild, 'Limitations of Carbon Footprint as Indicator of Environmental Sustainability', *Environmental science & technology*, vol. 46, pp. 4100–8, Mar. 2012, doi: 10.1021/es204163f.
[26] D. Peñaloza, M. Erlandsson, J. Berlin, M. Wålinder, and A. Falk, 'Future scenarios for climate mitigation of new construction in Sweden: Effects of different technological pathways', *Journal of Cleaner Production*, vol. 187, pp. 1025–1035, Jun. 2018, doi: 10.1016/j.jclepro.2018.03.285.
[27] A. Chapagain and P. Crovella, 'Building Sustainable Futures: Evaluating Embodied Carbon Emissions and Biogenic Carbon Storage in a Cross-Laminated Timber Wall and Floor (Honeycomb) Mass Timber Building', *Sustainability*, vol. 17, p. 5602, Jun. 2025, doi: 10.3390/su17125602.
[28] A. Dodoo, A. Al-Najjar, and J. A. Myhren, 'Environmental benefits of post-use materials recovery: a circular economy case study of a cross-laminated timber multi-storey building

in a life cycle perspective', in *World Conference on Timber Engineering 2025*, Brisbane, Australia: World Conference On Timber Engineering 2025, 2025, pp. 2288–2295. doi: 10.52202/080513-0279.
[29] A. Younis and A. Dodoo, 'Cross-laminated timber for building construction: A life-cycle-assessment overview', *Journal of Building Engineering*, vol. 52, p. 104482, Jul. 2022, doi: 10.1016/j.jobe.2022.104482.
[30] A. Dodoo, T. Nguyen, M. Dorn, A. Olsson, and T. K. Bader, 'Exploring the synergy between structural engineering design solutions and life cycle carbon footprint of cross-laminated timber in multi-storey buildings', *Wood Material Science & Engineering*, vol. 17, no. 1, pp. 30–42, Jan. 2022, doi: 10.1080/17480272.2021.1974937.
[31] S. Zhu and H. Feng, 'Enhancing circularity of wood waste through deconstruction in building sector', *Journal of Cleaner Production*, vol. 485, p. 144382, Dec. 2024, doi: 10.1016/j.jclepro.2024.144382.
[32] N. Ahn, A. Dodoo, M. Riggio, L. Muszynski, L. Schimleck, and M. Puettmann, 'Circular economy in mass timber construction: State-of-the-art, gaps and pressing research needs', *Journal of Building Engineering*, vol. 53, p. 104562, Aug. 2022, doi: 10.1016/j.jobe.2022.104562.
[33] A. Jayalath, S. Navaratnam, T. Ngo, P. Mendis, N. Hewson, and L. Aye, 'Life cycle performance of Cross Laminated Timber mid-rise residential buildings in Australia', *Energy and Buildings*, vol. 223, p. 110091, Sep. 2020, doi: 10.1016/j.enbuild.2020.110091.
[34] J. H. Andersen, N. L. Rasmussen, and M. W. Ryberg, 'Comparative life cycle assessment of cross laminated timber building and concrete building with special focus on biogenic carbon', *Energy and Buildings*, vol. 254, p. 111604, Jan. 2022, doi: 10.1016/j.enbuild.2021.111604.
[35] R. Passarelli, 'The Environmental Impact of Reused CLT Panels: Study of a Single-Storey Commercial Building In Japan', presented at the World Conference on Timber Engineering, Aug. 2018.
[36] R. Liu, L. Yao, Y. Gong, and Z. Wang, 'Life Cycle Assessment with Carbon Footprint Analysis in Glulam Buildings: A Review', *Buildings*, vol. 15, no. 12, Jun. 2025, doi: 10.3390/buildings15122127.
[37] E. Feys and E. Defever, 'Reusing glulam structures: Analysis of reclamation processes and environmental impact based on 7 recent cases', Sep. 2025.
[38] C. Piccardo and M. Hughes, 'Design strategies to increase the reuse of wood materials in buildings: Lessons from architectural practice', *Journal of Cleaner Production*, vol. 368, p. 133083, Sep. 2022, doi: 10.1016/j.jclepro.2022.133083.
[39] C. De Wolf, E. Hoxha, and C. Fivet, 'Comparison of environmental assessment methods when reusing building components: A case study', *Sustainable Cities and Society*, vol. 61, p. 102322, Oct. 2020, doi: 10.1016/j.scs.2020.102322.
[40] S. Tsoka and K. Tsikaloudaki, 'Design for Circularity, Design for Adaptability, Design for Disassembly', in *Circular Economy Design and Management in the Built Environment: A Critical Review of the State of the Art*, L. Bragança, P. Griffiths, R. Askar, A. Salles, V. Ungureanu, K. Tsikaloudaki, D. Bajare, G. Zsembinszki, and M. Cvetkovska, Eds, Cham: Springer Nature Switzerland, 2025, pp. 257–272. doi: 10.1007/978-3-031-73490-8_9.
[41] *Environmental management - Life cycle assessment - Principles and framework*, ISO 14040:2006, Geneva, Switzerland., 2006.
[42] *Sustainability of construction works - Assessment of environmental performance of buildings - Calculation method*, EN 15978:2011, Brussels, Belgium., 2011.
[43] M. Haupt and M. Zschokke, 'How can LCA support the circular economy?—63rd discussion forum on life cycle assessment, Zurich, Switzerland, November 30, 2016', *Int J Life Cycle Assess*, vol. 22, no. 5, pp. 832–837, May 2017, doi: 10.1007/s11367-017-1267-1.
[44] L. C. M. Eberhardt, H. Birgisdóttir, and M. Birkved, 'Life cycle assessment of a Danish office building designed for disassembly', *Building Research & Information*, vol. 47, no. 6, pp. 666–680, Aug. 2019, doi: 10.1080/09613218.2018.1517458.

[45] R. Garcia, R. A. F. Alvarenga, S. Huysveld, J. Dewulf, and K. Allacker, 'Accounting for biogenic carbon and end-of-life allocation in life cycle assessment of multi-output wood cascade systems', *Journal of Cleaner Production*, vol. 275, p. 122795, Dec. 2020, doi: 10.1016/j.jclepro.2020.122795.
[46] M. Brandão *et al.*, 'Key issues and options in accounting for carbon sequestration and temporary storage in life cycle assessment and carbon footprinting', *Int J Life Cycle Assess*, vol. 18, no. 1, pp. 230–240, Jan. 2013, doi: 10.1007/s11367-012-0451-6.
[47] Oslotre, 'HasleTre', Oslotre. Accessed: Feb. 07, 2026. [Online]. Available: https://oslotre.no/project/hasle-tre/
[48] Multiconsult, 'Climate gas calculation: Hasle Tre', RAP-RIM-02, Feb. 2022.
[49] O. Flindall, J. Tycho, and M. Groba, 'Hasle Tre: Norway's First Timber Office Building Designed for Disassembly and Reuse', presented at the World Conference on Timber Engineering, Oslo, Norway, Jun. 2023. doi: https://doi.org/10.52202/069179-0202.
[50] Granab, 'Granab Subfloor systems'. Accessed: May 13, 2026. [Online]. Available: https://www.granab.se/en/
[51] 'X-fix C 96-130-x', X-fix wood connector GmbH. Accessed: Apr. 16, 2026. [Online]. Available: https://x-fix.at/en/pages/x-fix-c-96-130-x
[52] dingmengzu, 'Hasletre / Save The Children Norway', IDPA Japan Design Award. Accessed: Apr. 17, 2026. [Online]. Available: https://idpa-japan.com/hasletre/
[53] 'Hasletre Office - Architectures - Jidipi'. Accessed: Apr. 17, 2026. [Online]. Available: https://architectures.jidipi.com/j00089564/en/hasletre-office
[54] 'Timber Design: Explained : Timber Development UK'. Accessed: May 20, 2026. [Online]. Available: https://timberdevelopment.uk/design/timber-design-explained/
[55] *Sustainability of construction works - Environmental product declarations - Core rules for the product category of construction products*, EN 15804:2012, Brussels, Belgium., 2012.
[56] European Commission. Joint Research Centre., *Model for Life Cycle Assessment (LCA) of buildings*. LU: Publications Office, 2018. doi: 10.2760/10016.
[57] A. Davis, A. Quintana-Gallardo, N. Martí Audí, and I. Guillén Guillamón, 'The impact of lifespan assumptions in LCA: Comparing the replacement of building parts versus building layers—A housing case study', *Energy and Buildings*, vol. 326, p. 115050, Jan. 2025, doi: 10.1016/j.enbuild.2024.115050.
[58] Z. Chen, H. Gu, R. D. Bergman, and S. Liang, 'Comparative Life-Cycle Assessment of a High-Rise Mass Timber Building with an Equivalent Reinforced Concrete Alternative Using the Athena Impact Estimator for Buildings', *Sustainability*, vol. 12, no. 11, Jun. 2020, doi: 10.3390/su12114708.
[59] L. Lin, D. Trabucco, and G. Perrucci, 'End-of-Life Scenarios for Mass Timber: Assumptions, Limitations and Potentials—A Literature Review', *Applied Sciences*, vol. 15, no. 3, p. 1208, Jan. 2025, doi: 10.3390/app15031208.
[60] 'CCBuild - Marketplace'. Accessed: May 08, 2026. [Online]. Available: https://ccbuild.se/en/marketplace/
[61] 'Sirkulær Ressurssentral'. Accessed: May 08, 2026. [Online]. Available: https://www.ressurssentral.no/prosjekter
[62] *Sustainability of construction works - Environmental product declarations - Core rules for the product category of construction products*, EN 15804:2012+A1:2013, Brussels, Belgium., 2013.
[63] *Sustainability of construction works - Environmental product declarations - Core rules for the product category of construction products*, EN 15804:2012+A2:2019, Brussels, Belgium., 2019.
[64] '"HasleTre" Oslo/Norway - X-fix in the 3000 m^2 Oslotre office building', X-fix wood connector GmbH. Accessed: Apr. 16, 2026. [Online]. Available: https://x-fix.at/en/blogs/news-1/hasletre-oslo-norwegen-x-fix-im-3000-m-grossen-burogebaude-von-oslotre

[65] X-fix wood connector GmbH, 'Technical dimensions - data sheet, for X-fix connectors', Austria, Feb. 2025.
[66] Simpson Strong-Tie, 'Technical Data Sheet: TTUFS Countersunk Wood Screw', Simpson Strong-Tie, C-F-EU-TG-2023–2, 2023. [Online]. Available: https://pim.strongtie.eu/api/v1/public/download/lv/en/product/1157/TTUFS.pdf
[67] R. Tomasi, Y. De Santis, A. Aloisio, R. Crocetti, V. H. Sæby, and A. Q. Nyrud, 'Experimental and analytical investigation on timber connections with beech, birch and laminated densified wooden dowels', *Construction and Building Materials*, vol. 494, p. 142692, Oct. 2025, doi: 10.1016/j.conbuildmat.2025.142692.
[68] Würth Group / Adolf Würth GmbH & Co. KG, 'Steel Screws (EPD-AWU-20230570-CBA1-EN)', Institut Bauen und Umwelt e.V. (IBU), Germany, EPD-AWU-20230570-CBA1-EN, Mar. 2024.
[69] 'VGS PLATE: The first reusable screw for lifting systems'. Accessed: Apr. 21, 2026. [Online]. Available: https://www.rothoblaas.com/blog/vgs-plate-the-first-reusable-screw-for-lifting-systems

Appendix A
Detailed list of materials

This appendix details the list of materials on which the life cycle inventory is based (Table 0.1).

Table 0.1: detailed list of materials of HasleTre, split into reusable and non-reusable materials

	Reusable: Sum of Mass of raw materials (kg)	**Non Reusable: Sum of Mass of raw materials (kg)**
215 - Pile foundations		
Ready-mixed concrete for foundations and internal walls	24692	
Ready-mixed concrete for external walls and floors		856
Structural steel and steel sections	37562	1290
222 - Columns		
Ready-mixed concrete for external walls and floors	27264	13200
Steel reinforcement for concrete (rebar)	3692	
Cross-laminated timber (CLT), glued laminated timber and laminated veneer lumber (LVL)	8087.76	
Reinforcing steel for concrete		825
Solid timber and glued laminated timber	17263.1	
223 - Beams		
Ready-mixed concrete for external walls and floors	494.4	
Steel reinforcement for concrete (rebar)	62	
Cross-laminated timber (CLT), glued laminated timber and laminated veneer lumber (LVL)	24489.82	
Solid timber and glued laminated timber	33600.3	
231 - Load-bearing external walls		
Ready-mixed concrete for external walls and floors		653736
Steel reinforcement for concrete (rebar)		15640
Glass wool insulation		938.75
Plywood		952.69
Cross-laminated timber (CLT), glued laminated timber and laminated veneer lumber (LVL)	82124.52	
Plastic membranes		313.16
Special plasterboards		7713.12
Rock wool insulation		602.65
Wood / sawn timber (softwood and hardwood)	23053.26	7133.28
Standard plasterboards		11575.04
Ecological insulation	1667.2	
Reinforcing steel for concrete		8700
Wood/sawn timber		2236.15
Organic insulation		3723.45
233 - Glass façades		
Aluminium	141	

	Reusable: Sum of Mass of raw materials (kg)	**Non Reusable: Sum of Mass of raw materials (kg)**
Flat glass	12560	
Wood / sawn timber (softwood and hardwood)	1884	
234 - Windows, doors, gates		
Aluminium	175	
Metal and industrial doors	1087.5	
Flat glass	15520	
Wood / sawn timber (softwood and hardwood)	2328	
Wooden doors		501.02
Glass doors		442.17
Glass and glass façades		15652
Wood-framed windows		10565.33
235 - External cladding and surfaces		
Paint, coatings and varnish	245.164	
Wood / sawn timber (softwood and hardwood)	12045.89	
Wood/sawn timber		7342.77
241 - Load-bearing internal walls		
Ready-mixed concrete for external walls and floors		60000
Glass wool insulation		3890.09
Cross-laminated timber (CLT), glued laminated timber and laminated veneer lumber (LVL)	27856.4	
Structural steel and steel sections		1715.54
Standard plasterboards		43970.73
Reinforcing steel for concrete		1875
Solid timber and glued laminated timber	43240	
Wood/sawn timber		2335.79
243 - System walls, glass panels		
Partition systems	12038.51	
System walls (windows)		9915.7
244 - Windows, doors, folding walls		
Wooden and timber-panel doors	6179.21	
System walls (windows)		4462.58
25 - Floors		
Plastic membranes		1148
251 - Self-supporting floors		
Ready-mixed concrete for external walls and floors		285600
Glass wool insulation		3806
Cross-laminated timber (CLT), glued laminated timber and laminated veneer lumber (LVL)	126266.45	
Special plasterboards		19376
Structural steel and steel sections	15229	
Standard plasterboards		37006.56
Ecological insulation		11564.55

	Reusable: Sum of Mass of raw materials (kg)	Non Reusable: Sum of Mass of raw materials (kg)
Reinforcing steel for concrete		11900
Solid timber and glued laminated timber	114595.2	
Raised floor systems		25188.8
252 - Ground floor		
Ready-mixed concrete for external walls and floors	1270152	393600
Steel reinforcement for concrete (rebar)	45017	
Reinforcing steel for concrete		16400
XPS (extruded polystyrene) insulation		3616
Sand, soil and gravel		147200
EPS (expanded polystyrene) insulation	211.2	
255 - Floor surface		
Carpet flooring		4243.25
Treated or coated timber		8082.45
Floor covering		4552.02
261 - Primary structure		
Cross-laminated timber (CLT), glued laminated timber and laminated veneer lumber (LVL)	49610	
Plastic membranes		115.81
Wood / sawn timber (softwood and hardwood)		5035.85
Solid timber and glued laminated timber	51832.8	
EPS (ekspandert polystyren) isolasjon		3004.8
EPS (expanded polystyrene) insulation		3092.16
281 - Internal staircases		
Ready-mixed concrete for external walls and floors	100800	7200
Steel reinforcement for concrete (rebar)	4182	
Reinforcing steel for concrete		300
Solid timber and glued laminated timber	3596.6	
21 - Ground and foundations		
Ready-mixed concrete for external walls and floors		45600
Reinforcing steel for concrete		2850
262 - Roofing		
Plastic membranes		1095.5
264 - Roof structure		
Wood/sawn timber		4379
Sand, soil and gravel		72500
Engineered aggregates		3552.5
Grand Total	2200845	2014113
Proportion reusable		52 %

Appendix B
Calculations of the quantity of connections

This appendix details the calculations made to estimate the quantity of connections in HasleTre Circular and HasleTre Reference.

1. HasleTre Circular

In HasleTre Circular, 600 X-fix (type C90) were used (Figure 0.1) [64]. With a mass of 0.6 kg per X-fix [65], the total mass was calculated as shown in Equation 0.1.

$$M = m \times n = 0.6 \times 600 = 360\ kg \qquad \textit{Equation 0.1}$$

Where M is the total mass of X-fix, m the mass of 1 X-fix and n is the number of X-fix.

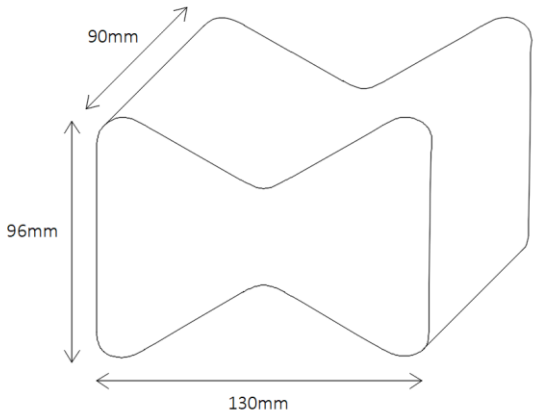


Figure 0.1: X-fix and its dimensions

In parallel, the number and size of beech dowels (Figure 0.2) was determined from the architectural plans and sections, enabling their total volume to be calculated as shown in Equation 0.2.

$$V = \pi \times r^2 \times L = \pi \times 12.5^2 \times 330 = 162\,000\ mm^3 = 0.000162\ m^3 \qquad \textit{Equation 0.2}$$
$$V_{tot} = V \times n = 0.000162 \times 222 = 0.036\ m^3$$

Where V is the volume of one dowel, r the radius of the dowel, L the length of the dowel, V_{tot} is the total volume of beech for all the dowels and n is the number of dowels.

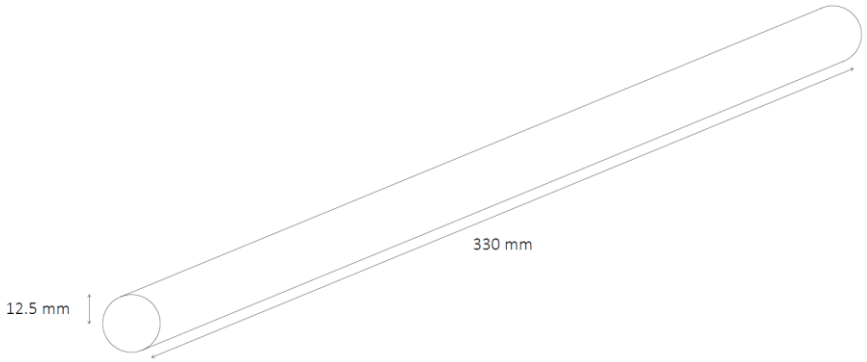


Figure 0.2: wood dowel and its dimensions

2. HasleTre Reference

The metal fasteners replacing the wooden connections are calculated as follows.

According to the manufacturer's data, one X-fix connector (type C90) can replace 24 screws with a diameter of 6 mm and a length of 80 mm [51]. To replace the 600 X-fix connectors in the project, the mass of metal required was calculated on the basis of a unit mass of 0.01 kg per screw [66], as shown below (Equation 0.3).

$$M = N \times m \times n = 24 \times 0.01 \times 600 = 144\ kg \qquad \textit{Equation 0.3}$$

Where M is the total mass of steel, N is the number of screws needed to replace one X-fix, m the mass of 1 screw and n is the number of X-fix.

As for the wooden dowels, these were replaced by steel dowels of 8 mm radius, the dimensions of which were estimated to provide equivalent structural capacity, based on the Johansen yield theory, as prescribed in Eurocode 5. This approach is supported by the methodology outlined by R. Tomasi et al. [67] in their study on timber connections. The mass of steel required is calculated in Equation 0.4.

$$V = \pi \times r^2 \times L = \pi \times 8^2 \times 330 = 66\,350\ mm^3 = 0.000066\ m^3 \qquad \textit{Equation 0.4}$$
$$M = n \times V \times \rho = 222 \times 0.000066 \times 7850 = 115\ kg$$

Where V is the volume of a steel dowel, r the radius of the steel dowel and L its length. M is the total mass of steel, n is the number of dowels and ρ the density of steel.

Their environmental impacts were calculated based on an EPD for steel screws (EPD-AWU-20230570-CBA1-EN) [68].

Appendix C
Assumptions in One Click LCA

Table 0.1 shows the assumptions regarding quantity, distance, wastage, service life, and EoL scenario for One Click LCA

Table 0.1: assumptions of One Click LCA per material

Material	Quantity (tonnes)	Distance of transport (km)	Method of transport	Wastage (%)	Service life (years)	EoL scenario
Aluminium	2.7	40	Trailer combination, 40-tonne capacity, 100 % fill rate	1	As building	Recycling and reuse
Reinforcement steel	111.4	110	Trailer combination, 40-tonne capacity, 100 % fill rate	5	As building	Recycling
Steel beams	55.8	110	Trailer combination, 40-tonne capacity, 100 % fill rate	1	As building	Recycling and reuse
Concrete	2883.2	70	Concrete mixer truck, appr. 8 m^3, 100 % fill rate	5	As building	Crushing to aggregate and reuse
Steel connections	0.26	110	Trailer combination, 40-tonne capacity, 100 % fill rate	1	As building	Recycling and reuse
CLT	530.9	130	Trailer combination, 40-tonne capacity, 100 % fill rate	10	As building	Reuse (and incineration)
Glulam	51.7	130	Trailer combination, 40-tonne capacity, 100 % fill rate	10	As building	Reuse (and incineration)
Plywood	0.9	130	Trailer combination, 40-tonne capacity, 100 % fill rate	10	As building	Incineration
Birch Plywood	0.3	130	Trailer combination, 40-tonne capacity, 100 % fill rate	10	As building	Reuse
Sawn timber	67.8	130	Trailer combination, 40-tonne capacity, 100 % fill rate	10	As building	Incineration and reuse
EPS	6.3	180	Trailer combination, 40-tonne capacity, 100 % fill rate	5	As building	Reuse and incineration
XPS	3.6	180	Trailer combination, 40-tonne capacity, 100 % fill rate	5	As building	Incineration
Wood fibre insulation	16.9	130	Trailer combination, 40-tonne capacity, 100 % fill rate	5	As building	Reuse
Stone wool insulation	0.6	70	Trailer combination, 40-tonne capacity, 100 % fill rate	5	As building	Landfilling
Glass wool insulation	8.6	70	Trailer combination, 40-tonne capacity, 100 % fill rate	5	As building	Landfilling
Gypsum plasterboard for floor	19.4	70	Trailer combination, 40-tonne capacity, 100 % fill rate	10	40	Recycling
Standard gypsum plasterboard	92.5	70	Trailer combination, 40-tonne capacity, 100 % fill rate	10	As building	Recycling

Gypsum plasterboard for external use	7.7	70	Trailer combination, 40-tonne capacity, 100 % fill rate	10	40	Recycling
Aggregate	150.7	70	Dumper truck, 19 tonne capacity, 100 % fill rate	1	As building	Backfilling
Soil	72.5	20	Dumper truck, 19 tonne capacity, 100 % fill rate	0	As building	Backfilling
Oil stain	0.25	150	Large delivery truck, 9 tonne capacity, 100 % fill rate	10	10	Reuse
Plastic vapour barrier	1.6	110	Trailer combination, 40-tonne capacity, 100 % fill rate	10	30	Incineration
Waterproofing membrane	1.1	110	Trailer combination, 40-tonne capacity, 100 % fill rate	10	30	Incineration
Solid wood cover	8.1	130	Trailer combination, 40-tonne capacity, 100 % fill rate	10	As building	Incineration
Linoleum	4.5	110	Trailer combination, 40-tonne capacity, 100 % fill rate	15	20	Landfilling
Carpet	4.2	110	Trailer combination, 40-tonne capacity, 100 % fill rate	10	15	Incineration
GRANAB subfloor system	25.2	110	Trailer combination, 40-tonne capacity, 100 % fill rate	5	25	Recycling
Balcony glass doors	0.4	60	Trailer combination, 40-tonne capacity, 100 % fill rate	0	40	Recycling
Industrial doors	1.1	130	Trailer combination, 40-tonne capacity, 100 % fill rate	0	30	Reuse
Wooden doors	6.7	130	Trailer combination, 40-tonne capacity, 100 % fill rate	0	40	Reuse and incineration
Windows	10.6	60	Trailer combination, 40-tonne capacity, 100 % fill rate	0	40	Recycling
Glass	41.4	60	Trailer combination, 40-tonne capacity, 100 % fill rate	1	35	Recycling and reuse
Wooden partition walls	1.2	70	Trailer combination, 40-tonne capacity, 100 % fill rate	10	As building	Reuse
Glass partition system	14.4	60	Trailer combination, 40-tonne capacity, 100 % fill rate	0	As building	Recycling

Appendix D

Ranking of technical solutions and systemic barriers in the Focus Group Discussion

This appendix provides an overview of the quantitative data collected during the FGD held on March 5, 2026. The definitions presented in this appendix are synthesised from the FGD. During the session, five industry experts participated in a digital Wooclap exercise to rank various technical solutions and identify the barriers to circularity in the Nordic timber construction sector, asked through the following questions. The final ranking is presented for each question.

What are the best technical solutions to enhance circularity of wood-frame buildings (downstream, production)?

1. **Modular element:** two-dimensional structural components, such as slabs, beams or pillars, that allow for repetition and standardisation.
2. **Reversible systems:** Mechanical fastening solutions designed for non-destructive disassembly. Examples discussed include X-fix wood connector for CLT alignment and shear forces, and reusable lifting screws like the VGS plate, which features a wear-indicator coating for safe multiple-use [51], [69].
3. **Non-load-bearing partitions:** Interior walls that can be reconfigured without affecting the primary structural integrity, extending the building utility as tenant needs change.
4. **Dry assembled wall layers:** Construction method that avoid foam and glue to ensure every component can be separated into parts for recycling or individual reuse.
5. **Volumetric design:** Three-dimensional self-supporting and prefabricated units that can be stacked to form a building.
6. **Wood dowels and wood nails:** timber-to-timber fasteners that allow be reworked with tools at the end-of-life without the risk of embedded metal.
7. **Accessible service voids:** Designing floor and ceiling systems with extra spatial allowances for HVAC and pipes to prevent structural obsolescence when technical systems are updated.

What do you see as the biggest obstacles when reusing structural materials from previous buildings?

1. Procurement
2. Missing value-chain forums
3. Missing marketplaces/business models
4. CE classification/guarantees
5. Regulations (BBR, etc.)
6. Technical solutions
7. Design process not adapted
8. LCA calculations
9. Warehouses for reused materials
10. Digital tools
11. Lack of good examples
12. Testing/reconditioning

What obstacles do you perceive for making future multi-story wood buildings more circular ?

1. Organisational/design processes not adapted
2. Laws and regulations

3. Procurement process
4. Missing value-chaine forums
5. Lack of good examples
6. LCA calculations
7. Lack of digital tools
8. Technical solutions